

General Industries Hub

Enterprise Inventory Management

Product Manual & User Catalogue

Version 2.0 - Generated 2026-06-17

Confidential - for authorized personnel only

GI Hub - Streamlit + SQLite + Twilio + Ollama - Multi-Site ERP

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General Industries Hub - Product Manual & User Catalogue

Version 3.0 - Multi-Site Warehouse Inventory ERP + Procurement Chain Document Scope: Complete operational reference for every role, page, tab, and element built into the system. Companion documents: handoff.md (technical architecture for engineers) - SOP.md (daily/weekly procedure for the Logistics + Warehouse teams).

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory ERP built on Streamlit + SQLite. It tracks every material movement (consumption, receipt, return, adjustment) through a two-stage approval ledger, enforcing real-time stock, FEFO (First-Expiry-First-Out) lot discipline, valuation, and audit trail across every site you manage.

1.2 Core principles

Principle	What it means
Identity math, not stored counters	$\text{Current_Stock} = \text{Total_Received} - \text{Total_Consumed} - \text{Total_Returned}$. The number is always computed from the immutable movement ledger - never written to a column. This makes drift impossible.
Two-stage approval	Store Keepers stage entries -> HOD reviews and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal request + approval.
Audit-first	Every consequential action writes to <code>system_audit_log</code> with username + timestamp + details. Even deletions leave a trace.
Self-healing schema	The DB layer automatically adds missing columns/tables on startup. You never need to run migrations manually.
Procurement chain (v3.0)	A SQL-driven workflow from Site PR through Logistics PO, Warehouse DN, and Site SK receipt. Every state transition is audited. Logistics owns POs; Warehouse owns physical receiving + DN preparati...
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.
Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see <code>Unit_Price</code> , <code>Total_Price</code> , or any monetary header field on a PO. Three independent enforcement layers guarantee this.

1.3 The transaction lifecycle (the heart of the system)


```

?????????????????????   ??????????????????????????   ??????????????????????????   ??????????????????????????
? Store Keeper   ?????? pending_issues   ?????? HOD EOD Review   ?????? consumption   ?
? enters issue   ?   ? (status='pending_   ?   ? (edit/approve/   ?   ? (permanent   ?
?               ?   ? hod')               ?   ? reject/commit)   ?   ? ledger)       ?
?????????????????????????   ??????????????????????????   ??????????????????????????   ??????????????????????????
                                     ?
                                     ?
                                     ??????????????????????????
                                     ? WhatsApp alerts   ?
                                     ? + audit log         ?
                                     ??????????????????????????

```

The same shape applies to receipts (pending_receipts -> receipts) and adjustments (stock_adjustments -> consumption/receipts).

1.3a The procurement chain (v3.0)

Site HOD	-> submits PR	-> Logistics: Incoming PRs
Logistics	-> issues PO	-> Site HOD: notifies + Admin: oversight
Logistics	-> assigns PO/items	-> Warehouse: Incoming Assignments
Warehouse user	-> acks + receives from vendor (records physical arrival)	
Warehouse user	-> drafts DN (RL/BL safe) -> Logistics: ?? DN approval queue	
Logistics	-> approves date	-> HOD: ? DN Approvals tab
HOD	-> approves content -> SK: ? Incoming DNs (Receipt Staging)	
Store Keeper	-> marks received -> receipts ledger + DN closed	

Side-paths: vendor returns (any role can raise), reschedules (Warehouse/HOD -> Logistics), force-closures (Logistics only, audited to Admin + Site HOD).

1.4 Currency, dates, units

- Currency: SAR (Saudi Riyal). All money is stored as REAL and displayed via format_sar().
- Dates: ISO format internally (YYYY-MM-DD); display format DD/MM/YYYY or DD MMM YYYY.
- Time zone: Server local time. Audit timestamps include seconds.
- Units of Measure (UOM): Per-item on the inventory master (e.g., Pcs, Box, Roll, Can, m, kg). No automatic UOM conversion - issue UOM = receipt UOM.

2. Roles, Permissions & Page Access

2.1 Role hierarchy

```
store_keeper (0) < warehouse_user (1) ? supervisor (1) < hod (2) < logistics (3) < admin (4)
```

The hierarchy is parallel, not strictly linear - warehouse_user and supervisor sit at the same numeric level but are scoped differently (one to a warehouse, the other to a site). Procurement-chain pages (Logistics Portal, Warehouse Portal) are EXACT-role-locked in addition to the hierarchy check, so a numerically higher role (e.g. Logistics) does NOT inherit access to a lower role's page (e.g. HOD Portal) just because the hierarchy says it could. The hierarchy lives in config.py:ROLE_HIERARCHY; the exact locks live in main.py:_EXACT_ROLE_PAGES.

2.2 Page access matrix

Page	Store Keeper	Warehouse User	Supervisor	HOD	Logistics	Admin
Live Dashboard	[X]	[OK]	[OK]	[OK]	[OK]	[OK]
Entry Log	[OK]	[X]	[X]	[X]	[X]	[X]
HOD Portal	[X]	[X]	[X]	[OK]	[X]	[X] (hidden - uses Admin Portal)
? Logistics Portal (NEW)	[X]	[X]	[X]	[X]	[OK]	[OK] (admin shadow)
? Warehouse Portal (NEW)	[X]	[OK]	[X]	[X]	[X]	[OK] (admin shadow, picks WH in sidebar)
Admin Portal	[X]	[X]	[X]	[X]	[X]	[OK]
Reports	[X]	[OK]	[OK]	[OK]	[OK]	[OK]

2.3 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Warehouse User (NEW)	Their own warehouse only - POs assigned to them, DNs they've prepared, items received from vendors. Tied to users.Warehouse_ID.
Supervisor	Their own site only - Reports, Live Dashboard, Burn Rate all site-locked.

Role	What they see
HOD	Their own site only - but they can REQUEST material from other sites (Cross-Site tab) and submit PRs to Logistics.
Logistics (NEW)	All sites globally for PRs and POs they manage. No site lock - they sit above the site boundary.
Admin	All sites + all warehouses globally - has the "All Sites" filter on every multi-site view; warehouse picker in sidebar when shadowing the Warehouse Portal.

2.4 Default seeded accounts

The first time the app starts, these accounts are created. Change the passwords immediately.

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

No default logistics or warehouse_user accounts are seeded. Both roles are strictly admin-created - go to Admin Portal -> ? Users -> Add user. This is intentional: the procurement chain has commercial visibility (Logistics sees prices, Warehouse routes inventory) and seeded credentials would be a security liability. When you create a warehouse_user, set their Warehouse_ID to one of the values from Admin Portal -> ?? Master DB Editor -> warehouses table - without it, the user lands on the Warehouse Portal and sees an error telling them to ask Admin.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- ? Sign In button - validates credentials via bcrypt; on success, populates `st.session_state["gi_user"]` and routes to the first allowed page.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read contr...
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below minimum (or "All levels adequate" if clear)
Bug/Feature reporting	Opens a dialog	See §3.4

Element	Purpose	Notes
Theme toggle	Dark / Light mode switch	Persists per-session
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
render_brand_header	Live Dashboard, Entry Log	Gold subtitle accent + current date
render_brand_header_hod	HOD Portal	Purple subtitle accent + current date
render_brand_header_admin	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
 - Critical (red) - force-closures, T-0 delivery reminders
 - Warning (amber) - T-1/T-2 reminders, reschedule requests
 - Success (green) - DN approved / received successfully
 - ? Info (blue) - PR submitted, PO issued, assignments
- Per-row ? Mark read - flips just that notification
- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope)

you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing "? Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if a notification helper errors, the badge silently shows 0 instead of crashing the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by toggles in config.WHATSAPP_TRIGGERS (see §9).

3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

4. Store Keeper Manual

The Store Keeper is the warehouse-floor operator. They see only the Entry Log page and the sidebar shell.

4.1 Pages visible

- Entry Log (only)

4.2 Entry Log - Tab structure

The Entry Log has four tabs:

1. Consumption Log - record material issued out
2. Receipt Staging - record material arriving in
3. Returnable Items - track tools/items temporarily issued
4. Stock Count - submit physical-count reconciliations

4.3 Entry Log -> Consumption Log

This is where every material consumed by site operations is recorded.

4.3.1 Top section: Bulk OCR upload (expander)

Upload Handwritten Consumption List (OCR) - for when you have a handwritten list to bulk-stage instead of typing each row.

Element	Purpose
Input method radio	Switch between "Image upload (vision AI)" and "Paste text"
File uploader	Upload PNG/JPG/JPEG/WEBP of handwritten list (vision model reads it)
? Extract rows from image button	Triggers vision AI extraction -> preview rows
Paste text area	Alternative - paste tab/comma/pipe-separated rows
? Parse pasted rows button	Parse the textarea
OCR Review grid	Editable preview of extracted rows with ambiguous-match pickers
Confirm All & Stage button	Pushes all reviewed rows into pending_issues

4.3.2 Mobile camera barcode/QR scanner (expander)

Barcode / QR Scanner - opens the browser camera and reads SAP codes off labels.

Element	Purpose
Camera feed	Live preview (requires camera permission)
Manual SAP code entry text box	Fallback if camera unavailable - also accepts pasted scanner output
Detected code display	Shows the last successfully scanned code in green

When a code is scanned, the material selectbox auto-populates so you don't have to search.

4.3.3 ? Scan / Add New Item to Queue (main expander, expanded by default)

This is the workhorse panel. Reading top to bottom:

A. Recently-scanned ring buffer

Element	Purpose
?? Recent: pills	Up to 5 quick-tap buttons showing your last 5 items (SAP code + truncated description). Tapping one auto-fills the material selectbox.

B. 1. Select Material

Element	Purpose
Search by SAP Code or Description selectbox	Type-ahead search across the inventory master. Format: [SAP_Code] Description.

When you pick a material, two cards appear automatically:

C. Item Snapshot card

A compact dark card showing:

- SAP code (gold monospace)
- Material description (bold)
- 3-stat strip:
- ? SITE STOCK (color-coded: red=empty, amber=below min, green=ok)
- 30-DAY BURN (units consumed last 30 days)
- DAILY RATE
- Inline 70×22 SVG sparkline - last 30 days consumption trend
- Status badge bottom-right (OK / Low / Below Min / Empty)

D. FEFO panel (lot suggestion)

If lots exist for this item at your site:

Element	Purpose
FEFO - First Expiry, First Out header (amber)	Banner
Per-lot row	Lot number (monospace) - ×qty (dim) - "Exp <date>" (amber if <90d, red if expired) - USE FIRST badge on the first row

The system will automatically attach the top-FEFO lot number to your consumption row when you

submit.

E. Pull from a different lot (FEFO override) - expander

Only renders when 2 or more open lots exist for this item at your site.

Element	Purpose
Lot to pull from instead selectbox	Default: "Keep FEFO suggestion". Other options: all other open lots with expiry + remaining qty shown
Reason for override text box (200 char max)	Mandatory. Min 5 characters to activate the override
Amber confirmation banner	Appears once override is active: "FEFO override active: pulling from <chosen> instead of <suggested>"

When activated:

- Your consumption row gets Lot_Number = chosen + FEFO_Override = reason
- Audit log entry FEFO_OVERRIDE is written with full context
- A WhatsApp alert is queued to the site HOD in real time

F. 2. Fill Entry Details (dynamic form)

Fields appear based on the pending_issues schema. Every required field is marked with *.

Field	Type	Purpose
Date	Date picker (default: today)	When the consumption happened
Quantity	Number input (min 0.1)	How much you're issuing. Above this field is a stock badge showing current site stock; a red [!] warning appears if you type mo...
Work_Type	Selectbox (sourced from system_settings.Work_Type)	Classification: PM Work, Breakdown, Project, Shutdown, Inspection, etc.
Issued_By	Text (auto-filled with your username)	Who handed it over
Issued_To	Text (smart-defaults from your last entry)	Recipient personnel/department
Tank_No	Text	Equipment tag if applicable
Serial_No	Text (optional)	Item serial number if tracked
PR_Number	Text (smart-default)	Linked purchase request if applicable
Remarks	Text (optional)	Free-form notes

Hidden side-effects:

- Site_ID is auto-set to your site
- Lot_Number is auto-set to FEFO top suggestion (or your override)
- status is set to 'draft' until you submit

G. [!] Override Expiry Warning (amber card with checkbox)

If the system detects a short-dated batch at your site for this item, it will hard-block the Add to Grid button until you check this box, confirming you've physically pulled from the expiring batch first.

H. Add to Grid ?? button

Validates that:

- A material is selected
- No required fields are empty
- Quantity ? current site stock (over-issue guard)
- Expiry override is checked when needed

On success: inserts a pending_issues row with status='draft', toasts [OK] Added to staging queue, reruns the page.

4.3.4 Staging Queue section

Element	Purpose
Staging Queue header + count badge	Shows how many draft rows you've accumulated
Data editor table	Edit any row inline before submitting
Save Draft Edits button	Persists edits in-place (rows stay as drafts)
Submit Grid to HOD button	Flips all draft rows to status='pending_hod' + queues a WhatsApp alert to the site HOD with the full item list

After submission, the rows are locked from your view - they appear in HOD Portal -> EOD Commit tab for approval.

4.4 Entry Log -> Receipt Staging

For when materials arrive at your site.

4.4.1 Upload Delivery Note (OCR) - expander

Identical lane structure to consumption OCR - upload an image of the delivery note OR paste text. Vision AI extracts rows for bulk staging.

4.4.2 ? Add Receipt to Queue - expander (expanded by default)**A. 1. Select Material**

Element	Purpose
Search by SAP Code or Description selectbox	Same type-ahead search as consumption

When picked, a blue-tinted info card shows:

- Mat Code

—

UOM

B. 2. Fill Receipt Details (dynamic form)

Field	Type	Purpose
Date	Date picker (today)	Delivery date
Quantity	Number input	Units received
Supplier	Text (optional)	Vendor name
Lot_Number	Text (optional)	If blank + Expiry_Date is set, system auto-generates LOT-YYYYMMDD-SAP
Expiry_Date	Date picker (optional)	Lot expiry - when set, triggers lot master entry
PR_Number	Text (optional)	Link to an open Purchase Request
Serial_No / Vehicle_No / DN_No / Pallet_No / Mob_From / Mob_To / Prepared_by / Driver_Name	Text (optional)	Logistics tracking fields
Remarks	Text	Free notes

C. Add to Receipt Queue ?? button

Validates: material picked, required fields populated. Inserts a pending_receipts row with status='draft'.

4.4.0 ? Incoming Delivery Notes from Warehouse (NEW in v3.0)

A new expander appears at the TOP of the Receipt Staging tab. It's only populated when the Warehouse has prepared a DN bound for your site AND your HOD has approved it. If empty, the expander shows:
"Nothing inbound. Logistics -> HOD-approved DNs will appear here for you to confirm physical receipt."

When DNs are inbound, each appears as its own container:

Element	Purpose
DN header line	DN <number> - PO <number> - Warehouse <id> - DN Date <date>
Line count + total qty	At-a-glance: how many SKUs, how many units total
View lines expander	Material_Code, Description, Qty, UOM, Lot_Number, Expiry_Date, Remarks - read-only preview of every line
[OK] Mark as Received button	When you confirm physical receipt: writes one receipts row per DN line (with DN_Number, Warehouse_ID, PO_Number_Source for full traceback), flips the DN to received, and clears it from your list....

When to use:

- The Warehouse delivered the materials physically to your site
- HOD already approved the DN content (you'll see it appear without any action)
- You've inspected the goods and they match the DN qty + lot

When NOT to use:

-

Direct deliveries from supplier to your site (those go through the existing Add Receipt to Queue -> Submit to HOD flow below; the email-driven PR/PO path remains supported)

- Partial receipts (the current flow assumes you confirm the full DN qty - for partials, raise a Vendor Return on the diff after confirming and ask your HOD)

This is the FINAL step in the procurement chain. After you click Mark as Received:

- Logistics sees `dn_received_by_sk` in their notifications
- Warehouse sees the same
- The PO and PR move toward closure if this DN completes the order

4.4.3 Receipt Draft Queue section

Element	Purpose
Receipt Draft Queue header + count badge	Number of drafts you've accumulated
Data editor table	Inline edit before submit
Save Draft Edits button	Persists
Submit to HOD for Approval button	Flips drafts to status='pending_hod' + WhatsApp alert to HOD with item list

4.5 Entry Log -> Returnable Items

For tools, gauges, fittings temporarily handed to personnel - items that should come back.

4.5.0 Smart Scan workflow (new - 2026-06)

A two-step camera flow sits at the top of the tab. Use it to skip typing borrower + tool details by hand; the existing manual form still works exactly as before and stays visible underneath.

Step 1 - Scan the employee badge

1. Click into the Smart Scan (Beta) expander.
2. Hold the borrower's printed badge in front of the camera. (Badges are generated in Admin Portal -> Employees -> Roster + Badges - see §7.6.)
3. The system decodes the QR (which carries only the ID_Number, no PII), looks it up in the Employees master, and shows:
 - [OK] Green success card - EMP-1042 - Ahmed Ali (+9665...) - the borrower's name + phone are now staged.
 - [X] Red error card - Badge couldn't be read OR the employee is inactive/suspended. Re-snap or use the manual form.

Step 2 - Scan the tool

1. After a successful badge scan, a second camera input appears. Point it at the tool.
2. The active YOLO model (managed in Admin Portal -> ?? Tool Catalogue, see also docs/cv_training_guide.md for how the model gets trained and promoted) classifies the image. Three outcomes:

—

Confidence ? 0.75 -> [OK] Auto-fill. The detected tool name + borrower details flow into the manual form below; review and click "Issue Item".

- Confidence 0.30-0.74 -> [!] A "Top candidates" radio appears. Pick the right one and click "Use this tool". The form pre-fills.
- No active CV model OR confidence < 0.30 -> An info banner explains the fallback. Borrower fields stay pre-filled; type the tool name into the manual form yourself.

Click "Start over (clear scan)" any time to reset and scan a different borrower.

What happens if the AI is wrong

You're never locked in. Every Smart Scan write-through goes into the *manual* form's fields - Name, Phone, Material name. Edit anything that's wrong before clicking "Issue Item". The submitted record stores cv_detected=1 and cv_confidence so admin reports can later show adoption + accuracy telemetry. Manually-corrected loans don't get tagged as CV - that's intentional, see §10 for the data model details.

Automatic WhatsApp reminders (Phase 6E)

Once a loan is in the system, the WhatsApp worker fires four reminders automatically. The cadence and recipient list is:

Offset	Severity	In-app	WhatsApp
T?2h (2 hours before due)	info	Site SK badge	Borrower
T?0 (due now)	warning	Site SK badge	Borrower
T+2h (2 hours overdue)	warning	Site SK badge	Borrower + every Site SK
T+24h (24 hours overdue)	critical	Site SK + Supervisor badges	Borrower + every Site SK + every Site Supervisor

The borrower's phone is resolved first from the Employees master (CV-scanned loans) and falls back to the manually-typed borrower_phone column for legacy/manual loans. The system de-dupes per loan + offset so restarting the worker mid-day never causes double-fires. Admins can mute any of the four events via config.WHATSAPP_TRIGGERS if a particular escalation gets too noisy.

4.5.1 ? Issue a Returnable Item - expander (expanded by default)

Field	Type	Purpose
Material / Tool Name	Text	What you're handing out
UOM	Text (e.g., Pcs, Set)	Unit
Quantity	Number (min 0.1)	How many
Borrower Name	Text	Person taking custody
Borrower WhatsApp Number	Text (optional, +966 ...)	For overdue alerts
Expected Return Date	Date (default: tomorrow)	When you expect it back
Expected Return Time	Selectbox: 04:15 PM, 06:15 PM, Custom Time...	Time-of-day expectation
Issue Item button	Validates name+borrower, writes to	

returnable_items

4.5.2 Currently Borrowed Items section

Shows all borrowed (not yet returned) items at your site.

- Overdue red banner - when one or more items are past their expected return time
- Borrowed items table (styled HTML) with columns: ID, Material, UOM, Qty, Borrower, Phone, Given Time, Expected Return, Status (pill: green "On Loan" / red "[!] Overdue")
- Mark as Returned section:
- Selectbox of borrowed items
- [OK] Mark as Returned button - updates status='returned', busts cache, toast confirms

4.6 Entry Log -> Stock Count

For reconciling physical shelf count with system stock when they don't match (damage, expiry disposal, miscount, found-extra).

4.6.1 1. Select Material to Count

Element	Purpose
Search by SAP Code or Description selectbox	Pick the material to reconcile
Help caption	Explains when to use this tab

4.6.2 2. Enter Count Details (only renders after picking a material)

Left column

Element	Purpose
System Qty card	Read-only dashboard showing current stock at your site (auto-fetched)
? Counted Qty input	What you actually count on the shelf right now (defaults to system qty)
Variance preview banner	Color-coded: green "? Found N extra" or red "? Short by N" or grey "No variance"

Right column

Element	Purpose
Reason Code selectbox	9 options. Default depends on variance direction.
Notes textarea (300 char max)	Free explanation: "found behind shelf 3", "damaged in transit", etc.

Reason codes:

Code	Label
cycle_count	Cycle count correction
damaged	? Damaged / unusable
expired_disposal	? Expired - disposed

Code	Label
miscount_in	? Miscount - found extra
miscount_out	? Miscount - short
lost	? Lost / unaccounted
theft	? Suspected theft
return_to_supplier	<-? Returned to supplier
other	? Other (see notes)

4.6.3 Submit section

Element	Purpose
Amber warning banner	"Submitting this sends the count to your HOD for approval. No stock changes until they approve."
Submit Count for HOD Approval button	Disabled while variance = 0. On success: writes to stock_adjustments with status='pending_hod', queues WhatsApp to site HOD with count details, audit-logs.

4.6.4 Recent Adjustments at This Site (history)

Read-only table showing: ID, SAP Code, Material Name, Variance, Reason, Status, Submitted By, Submitted At.

4.7 Store Keeper - Use Cases

Use Case 1: Issue 10 Pipe Gaskets to PM Work

1. Log in -> Entry Log -> Consumption Log tab
2. Open ? Scan / Add New Item to Queue
3. Search "Pipe Gasket" -> pick from dropdown
4. Read the stock badge (e.g., " 156 Pcs"), confirm FEFO lot suggestion
5. Quantity: 10
6. Work_Type: PM Work
7. Issued_To: name of recipient
8. Tank_No / PR_Number: as applicable
9. Click Add to Grid ??
10. Repeat for any other items in this batch
11. Review the Staging Queue table -> edit if needed -> click Submit Grid to HOD

The row(s) are now in HOD's EOD queue. Stock won't decrease until HOD commits EOD.

Use Case 2: Receive a delivery of 200 units with expiry date

1. Entry Log -> Receipt Staging tab
2. Open ? Add Receipt to Queue

3.

Pick material, fill Quantity (200), Supplier, Expiry_Date (e.g., 2027-06-11)

4. Lot_Number: leave blank (system will generate LOT-20260611-<SAP>) OR type the supplier's lot ID
5. PR_Number: link if applicable
6. Click Add to Receipt Queue ??
7. Click Submit to HOD for Approval when done staging

Use Case 3: Found 5 extra units of an item (physical count > system)

1. Entry Log -> Stock Count tab
2. Pick the item from the dropdown
3. Read the System Qty card (e.g., "95")
4. Counted Qty: 100
5. The variance banner turns green: "? Found 5 extra"
6. Reason: ? Miscount - found extra
7. Notes: e.g., "Found in box behind shelf 3"
8. Click Submit Count for HOD Approval

HOD will approve in HOD Portal -> Adjustments. After approval, your live stock matches reality and a synthetic receipt row of +5 is posted to the ledger.

Use Case 4: Issue a tool to a borrower

1. Entry Log -> Returnable Items tab
2. ? Issue a Returnable Item
3. Material: Torque Wrench 1/2", UOM: Pcs, Qty: 1
4. Borrower Name, WhatsApp Number, Expected Return: today 06:15 PM
5. Click Issue Item

If the item is not returned by 06:15 PM, the borrower (and you) get a WhatsApp alert.

Use Case 5: FEFO override (the right bin is blocked)

1. Entry Log -> Consumption Log -> pick material with 2+ open lots
2. Scroll to FEFO panel - note the suggested lot (earliest expiry)
3. Open Pull from a different lot expander
4. Pick a different lot from the dropdown
5. Type a reason ? 5 chars (e.g., "FEFO bin blocked by pallet, clearing 1700")
6. Amber confirmation banner appears
7. Fill quantity + other fields
8. Click Add to Grid ??

Your HOD gets a real-time WhatsApp alert documenting the override.

4.8 Store Keeper - FAQ

Q: I can't see Live Dashboard, HOD Portal, or Reports. Did I lose access? A: No - Store Keepers only have access to Entry Log by design. If you need to view stock, ask your Supervisor or HOD.

Q: I submitted to HOD by mistake. Can I cancel? A: Not directly. Contact your HOD - they can reject the row in their EOD Commit review.

Q: The "Add to Grid" button is disabled or the form rejects my entry. A: Check three things: (a) is quantity > current stock? (b) is the expiry override needed but unchecked? (c) are all required * fields filled?

Q: I tried to issue 20 but the system says only 15 available. A: Believe the system. Either physically recount, or if there is genuinely 20 and the system disagrees, use the Stock Count tab to log a +5 adjustment.

Q: I don't have a barcode scanner. How do I quick-find items? A: Type a partial SAP code or description in the selectbox - it has type-ahead search. Or use the Recently-scanned pills if you've used the item recently.

Q: WhatsApp alerts aren't reaching the HOD when I submit. A: HOD's phone number must be set in their user profile. Tell the HOD to update it via Admin (or check with Admin directly).

Q: My borrowed item is overdue but I haven't received an alert. A: WhatsApp alerts are sent by a background worker (whatsapp_worker.py) - check with Admin that it's running.

Q: I uploaded an OCR image and it extracted wrong items. A: The OCR review grid lets you fix every row before staging. Pick the correct material from the suggested-matches dropdown, or type it manually.

Q: How do I see my recent activity? A: Submitted/approved items don't show on your screen. Ask HOD or Admin to filter the audit log by your username.

5. Supervisor Manual

The Supervisor monitors a single site's stock, generates reports, and provides oversight. They cannot approve transactions (that's HOD).

5.1 Pages visible

- Live Dashboard
- Entry Log (same interface as Store Keeper - see §4)
- Reports

5.2 Live Dashboard

The Live Dashboard is the at-a-glance view of every catalogue item with current stock, value, and status.

5.2.1 Brand header

`render_brand_header("Live Warehouse Stock Dashboard")` - gold subtitle + today's date.

5.2.2 Hero metric strip (4 cards)

Card	Source	Tone logic
Catalogue items	Count of rows in inventory	Neutral
Total stock value	<code>cached_total_inventory_value()</code> (Sum of <code>Current_Stock × Unit_Cost</code> , all sites) - formatted as SAR 1,234 / SAR 1.2M	Neutral. Delta: "standard cost - all sites" or "set Unit_Cost in Admin -> DB Editor" if 0
Below minimum	Count from <code>cached_low_stock_items()</code>	Green=0, Amber<10, Red>=10. Delta: "all healthy" or "needs reorder"
Expiring / expired	Count from <code>cached_short_dated_stock()</code>	Green=0, Amber<10, Red>=10. Delta: "shelf-life clear" or "review HOD Portal"

5.2.3 Ask in plain English (AI search) - expander (if AI_ENABLED)

If Ollama is running locally with the qwen2.5-coder:7b model:

Element	Purpose
Your question text input	Natural language query, e.g., "items below minimum at site B"
Search button	Translates to SQL via local LLM, runs read-only
Clear button	Resets the panel
Result table	Returned rows

Element	Purpose
"Show SQL the AI generated" expander	Shows the safe SELECT the LLM produced

If Ollama is NOT running: an amber card displays setup instructions (ollama serve).

5.2.4 Burn alert banner

Appears at the top of the table area when there are items burning to zero within the configured window (default 7 days). Color-coded amber/red.

5.2.5 Main inventory grid (AgGrid)

Column	Source	Notes
SAP_Code	inventory.SAP_Code	Primary key
Equipment_Description	inventory.Equipment_Description	Material name
UOM	inventory.UOM	Unit of measure
Total_Returned	Computed	Sum from returns table
Current_Stock	Computed identity math	Received - Consumed - Returned
Minimum_Qty	inventory.Minimum_Qty	Reorder threshold
Unit_Cost	inventory.Unit_Cost	Set via Admin -> DB Editor
Stock_Value	Computed	Current_Stock × Unit_Cost, rounded 2 dp
Status	Computed badge	OK / Low / Below Min / Empty with colored pill via STATUS_BADGE_JS

You can sort by any column - sorting by Stock_Value shows your biggest SAR exposure first.

5.2.6 Expanders below the grid

Expander	Content
Stock vs Minimum Threshold	Horizontal bar chart, sorted by criticality. Each bar = item; gold line markers = minimum threshold. Color: red=empty, amber=below min, orange=close, green=ok. Capped at 20 items.
Burn Rate Forecast (30-Day)	Plotly chart with daily-burn-rate bars + vertical line at the 30-day alert threshold. Colors: red <10 days, amber <30, green >30.
Top Consumed Items	Bar chart of top 10 items by 30-day consumption, blue->gold gradient.

5.3 Reports

Supervisor sees the full Reports module but with site scope locked to their site (cannot pick "All Sites"). See §8 for the full Reports reference.

5.4 Supervisor - Use Cases

Use Case 1: Daily morning check

1. Log in -> Live Dashboard
2. Read the 4 hero cards - note any below-minimum or expiring items
3. Sort grid by Stock_Value descending -> check no over-stocked items
4. Sort by Status -> spot Empty / Below Min items
5. Open Burn Rate Forecast expander -> check the next 30 days

Use Case 2: Generate end-of-month report for management

1. Reports -> Generate Report
2. Pick ? Monthly Summary
3. From date: 1st of month; To: today
4. Site: locked to your site
5. Format: PDF
6. Click ? Generate Report
7. Review the preview (includes SAR-value columns: Issued_Value_SAR, Received_Value_SAR, Closing_Value_SAR)
8. Click v Download PDF

Use Case 3: Investigate why an item ran out

1. Live Dashboard -> search the item (or click its row)
2. Open Burn Rate expander - see the daily rate trend
3. Reports -> Burn Rate Analysis -> date range last 30 days -> generate
4. Filter the result table for the SAP code -> see daily breakdown
5. Reports -> Daily Consumption -> narrow down which Work_Type drove the spike

5.5 Supervisor - FAQ

Q: I can only see my own site. Why can't I see other sites' stock? A: By design - site isolation. Only Admin sees all sites. If you need to know another site's stock for a transfer, ask your HOD to file a Cross-Site Request.

Q: Total stock value shows SAR 0. What's wrong? A: Items have no Unit_Cost set. Ask Admin to enter costs in Admin -> Master DB Editor -> inventory table.

Q: AI search says "Ollama not running." A: Local AI is optional. Admin must install Ollama and pull the qwen2.5-coder:7b model. Without it, the system still works - you just type into the standard grid filters.

Q: The Stock_Value column is empty for some items. A: Those items have Unit_Cost = 0. The valuation is correct - they're tracked in qty but not in money.

Q: I can't approve any transactions. A: Correct. Supervisors monitor; HODs approve. Talk to your HOD.

Q: Can I export the Live Dashboard grid? A: Yes - AgGrid has a built-in CSV export (right-click the grid). For a formal report, use Reports -> Generate.

6. HOD (Head of Department) Manual

The HOD owns their site's inventory ledger. Every transaction flows through their approval. The HOD Portal has 15 tabs as of v3.0 - the original 13 (covering EOD, Cross-Site, Burn Rate, Receipts, Returns, Adjustments, PRs, Shelf-Life, Notifications, My Requests, Site Config, DOC, QR Approval) plus two new procurement-chain tabs: ? DN Approvals and ? In-Transit. Existing tabs are unchanged; the new tabs are documented in §6.15 and §6.16 below.

6.1 Pages visible

- Live Dashboard (see §5.2)
- Entry Log (see §4 - HODs can also stage entries)
- HOD Portal - detailed below
- Reports (see §8)

6.2 HOD Portal overview

6.2.1 Page header

`render_brand_header_hod("HOD Management Portal")` - purple subtitle + today's date.

- Page title: " HOD Portal" + "Managing Site: <SITE_ID>"

6.2.2 Hero metric strip (4 cards)

Card	Source	Notes
Site stock value	<code>cached_total_inventory_value(site_id)</code>	Delta: <SAR> consumed (30d) - site-scoped
Below minimum (site)	<code>cached_low_stock_items(site_id)</code> count	Green/amber/red
Expiring / expired	<code>cached_short_dated_stock(site_id)</code> count	Green/amber/red
Pending receipts to approve	Count from <code>pending_receipts</code> where <code>status='pending_hod'</code> at your site	Neutral or amber

6.2.3 Tab strip (10 tabs)

1. EOD Commit
2. Cross-Site
3. Burn Rate
4. ? Pending Receipts
5. Adjustments
6. Purchase Requests

7.

Receive Material

8. [!] Shelf-Life

9. Notifications

10. [OK] My Requests

6.3 HOD Portal -> EOD Commit

The single most consequential action in the system: committing the day's staged consumption to the permanent ledger.

6.3.1 Top section

Element	Purpose
Header	" End-of-Day Commit (<SITE>)" with intro caption

6.3.2 4-card stat strip

Card	Source
Total entries	Total pending rows at your site
? Pending	Status = pending
[!] Flagged	Status = flagged
[OK] Approved	Status = approved

6.3.3 Filter pills

Pill	Action
All / Pending / Flagged / Approved / Rejected	Filters the table below to that status

6.3.4 Top action bar

Button	Action
[OK] Approve All Pending	Sets every pending row to status='approved'. Disabled when count = 0.
Commit EOD to Master	Opens the type-COMMIT modal. See §6.3.7

6.3.5 Main pending-issues table

Custom dark-themed HTML table showing every column from the staging row (joined with inventory for descriptions):

Column	Source
Date	pending_issues.Date
SAP	pending_issues.SAP_Code
Mat Code	inventory.Material_Code
Material	inventory.Equipment_Description
UOM	inventory.UOM

Column	Source
Qty	pending_issues.Quantity
Work Type	pending_issues.Work_Type
PR	pending_issues.PR_Number
Tank	pending_issues.Tank_No
Serial	pending_issues.Serial_No
Issued By	pending_issues.Issued_By
Issued To	pending_issues.Issued_To
Remarks	pending_issues.Remarks
Status	Colored pill (pending / flagged / approved / rejected / committed)

The table also reflects Lot_Number and FEFO_Override columns when present (visible by editing the row).

6.3.6 Row Actions panel (per-row Approve/Reject)

Below the table, top 20 actionable rows render as cards:

Element	Purpose
Card line 1	SAP - Material Code - Material name - Qty UOM
Card line 2	Date - Work Type - PR - Tank - Serial - By - To - Remarks (only fields with values)
[OK] button	Approve this single row (sets status='approved')
[X] button	Reject this single row (sets status='rejected')

6.3.7 EOD Commit confirmation modal (type-COMMIT dialog)

When you click Commit EOD to Master:

Step 1 - pre-flight: negative-stock guard

The system calls `validate_eod_no_negative_stock()`. If any items would push stock below zero:

Element	Purpose
? Red error banner	"Cannot commit - N item(s) would go negative"
Violation table	SAP - Material - Current Stock - Trying to Consume - Deficit
? Hint banner	"Common fixes: receive inbound stock first, raise a stock adjustment, or reduce the consumption to fit available stock."
Close button	Cancels - fix the staging rows and try again

Step 2 - confirm (only if no violations)

Element	Purpose
Warning banner	"You are about to commit N pending row(s)..."
Type COMMIT text input	Must type exactly COMMIT to enable the button
Cancel button	Abort

Element	Purpose
Confirm Commit button (red)	Calls commit_eod() - moves rows to consumption, deletes from pending_issues, busts caches, queues a post-EOD low-stock alert to the HOD if applicable.

After commit: ? balloons animation + success toast.

6.3.8 Flagged-items banner

If there are flagged rows: amber banner appears below the table reminding the HOD to verify with the store keeper before committing.

6.4 HOD Portal -> Cross-Site

For requesting material from another branch.

6.4.1 Top section

Element	Purpose
Intro caption	Explains the > 5-item escalation rule

6.4.2 Single-target inquiry flow (left/right columns)

Left column:

Element	Purpose
Select Target Branch selectbox	List of all sites except yours
Select Material selectbox	Type-ahead search across inventory
Quantity Needed number input	Min 1
Justification / Notes textarea	Why you need this transfer

Right column:

Element	Purpose
Live Stock at <target> header	Shows availability at the target site
Available Quantity metric	Live stock at target
Suggested Transfer Qty metric	min(requested, available)
? Add to List button	Adds to the cart

6.4.3 ? Your Request Cart section

Element	Purpose
Cart table (styled HTML)	Columns: Target Site - SAP - Description - Qty - Notes
Submit All Requests to Admin button	Creates a requests row per cart item, queues WhatsApp to all admins, AND (if >5 items) escalates to the target-site

HOD via separate WhatsApp + audit log

Element	Purpose
? Clear List button	Empties the cart

6.4.4 Incoming Cross-Site Requests panel (bottom)

Where a HOD sees requests addressed to their site from other HODs.

Element	Purpose
Intro caption	"Requests addressed to <site>"
Incoming table (styled HTML)	From - SAP - Material - Qty - By - Notes - When
Per-row Approve / Reject controls (top 10)	[OK] Approve and [X] Reject buttons
Approval action	Calls update_request_status(status='approved') + audit
Rejection action	Same with status='rejected'

6.5 HOD Portal -> Burn Rate

Predictive analysis of which items will run out and when.

6.5.1 Compact horizontal-bar chart (top 10)

A sleek card with:

- Header row: "Monthly Consumption - Top 10" + legend chips (red=High / amber=Mid / green=Low)
- Per-row line: material name (truncated) - gradient bar (color reflects burn intensity proportional to max) - monthly figure with UOM - days-remaining badge (red if ?7d, amber if ?30d, green if >30d)

6.5.2 Burn alert banner (above the Plotly chart)

Auto-renders if items are projected to hit zero within the alert window.

6.5.3 Full Plotly burn rate chart

Daily-burn-rate bars with vertical line at 30-day alert threshold. Color-coded.

6.5.4 Detailed Forecast Table (AgGrid)

Columns: SAP_Code - Equipment_Description - UOM - Current_Stock - Daily_Burn_Rate - Days_Remaining

Sortable + exportable.

6.6 HOD Portal -> ? Pending Receipts

Approve or reject receipts staged by Store Keepers.

6.6.1 Header + count banner

Amber banner: "? N pending receipt(s) require your approval before stock levels update."

6.6.2 [OK] Approve All button

Bulk: calls `commit_pending_receipts(site_id, username)` which:

1. Iterates every pending receipt
2. Calls `process_receipt_delivery` (which creates lots if `Expiry_Date` set)
3. Posts to receipts table
4. Auto-closes PR if fulfilled
5. Deletes the staged rows
6. Busts caches, audit-logs, success toast

6.6.3 Receipt table

Columns: Date - SAP - Equipment_Description - UOM - Quantity - Supplier - PR_Number - Site_ID - Status pill

6.6.4 Per-row Reject controls (top 10)

Element	Purpose
Card line	SAP - Qty UOM - Supplier
[X] Reject button	Soft-reject (row stays as status='rejected'), preserves audit, audit-logs with reason

6.7 HOD Portal -> Adjustments

Approve physical-count reconciliations submitted by Store Keepers from the Entry Log -> Stock Count tab.

6.7.1 3-card stat strip

Card	Source
? Pending	Count of pending adjustments at your site
? Shortfalls	Variance < 0 count
? Surpluses	Variance > 0 count

6.7.2 Pending adjustments table

Columns: # - SAP - Material - UOM - System - Counted - Variance (red for negative, green for positive) - Reason - By - When

6.7.3 Per-row Approve/Reject

Below the table:

Element	Purpose
Card line 1	SAP - Material - System X -> Counted Y - Variance
Card line 2	Reason label - submitted by - notes
[OK] button	Calls approve_stock_adjustment(adj_id, approver). Atomically posts a synthetic ledger row - if variance < 0, inserts into consumption with Work_Type=STOCK_ADJUSTMENT; if > 0, inserts into
[X] button	Reject. Row stays as status='rejected' for audit.

6.7.4 Recent Adjustments History (last 30)

Columns: # - SAP - Material - Variance - Reason - Status pill - Submitted By - Approved By - Ledger Ref

6.8 HOD Portal -> Purchase Requests

Manage Purchase Requests (PRs) for your site.

6.8.1 ? Create New PR (manual entry) - expander

Field	Purpose
PR Number	Free text, required (e.g., 3001234567)
Material selectbox	Picks from inventory; auto-fills Material_Code, Material_Name, UOM via blue-tinted info card
Requested Qty	Min 0.01
Preferred Supplier	Optional
Estimated Cost (SAR)	Optional
UOM	Auto-filled, editable
Notes	Optional
Create PR Draft button	Calls insert_manual_pr(...). Sets status='open', workflow_state='draft'

6.8.2 ? Upload PR PDF (auto-extract via pdfplumber) - expander

Element	Purpose
File uploader (PDF only)	Upload Purchase Request PDF
Process Upload button	Calls process_pr_pdf() - extracts PR number + Material Codes (GI-XXXXXXX pattern), matches to inventory's Material_Code, inserts pr_master rows
Result banner	Green success or amber warning (if some materials unmatched)

6.8.3 Current PRs table

Columns: PR No. - SAP - Material - UOM - Qty Req - Qty Pending - Supplier - Est. SAR - Workflow pill

Pending_Qty = Requested_Qty ? SUM(receipts where PR_Number matches)

6.8.4a ? Submit PR(s) to Logistics Portal (NEW in v3.0)

A new expander sits between the PR list and the email/PDF block. It opens the procurement-chain path - handing off your PR to the in-app Logistics queue instead of (or in addition to) the legacy email path.

Element	Purpose
Multi-select	All open PRs at your site that haven't yet been submitted to Logistics
Submit Selected to Logistics button	Calls submit_pr_to_logistics() for each picked PR. Their logistics_status flips to submitted and the row appears in Logistics Portal -> Incoming PRs. A pr_submitted_to_logistics notification fir...

The legacy email path (§6.8.4) still works - your team can use it for direct-to-vendor relationships that don't go through central Logistics. Both paths coexist. The email path is marked for future deprecation once procurement chain adoption reaches the agreed threshold.

6.8.4 ? Notify Logistics section

When at least one PR has status='open':

Element	Purpose
Select PR for Actions selectbox	Pick from open PRs
? Draft Outlook Email button	Opens email client with HTML table of pending items. Mac: opens Mail.app via AppleScript with formatted HTML table. Windows: opens Outlook via COM with HTMLBody. Fallback: mailto: with monospace ...
Download PR PDF button	Generates a PDF record with columns: Material Code - Description - Req. Qty - Received - Pending - Status. Auto-saved metadata: Site, Generated By, Timestamp.

6.9 HOD Portal -> Receive Material

For HOD-direct receipts (when a HOD logs a delivery themselves, e.g., direct purchases or HQ deliveries).

6.9.1 PR linker

Element	Purpose
? Link to Open PR selectbox	Filters the material list to items in that PR, or "None (Direct Purchase)" for free-form

6.9.2 Receipt form (st.form, clears on submit)

Field	Type	Purpose
Select Material	Selectbox	From inventory or filtered to PR items
Quantity Received	Number (min 0.1)	Units arrived
Delivery Date	Date (today)	When
Expiry Date	Date (optional)	Lot expiry trigger
Logistics extras	Text inputs	All non-system columns on receipts (Vehicle_No, Driver_Name, DN_No, Supplier, Remarks, etc.)
Save Receipt button	Calls process_receipt_delivery() directly (no staging) - instantly posts to receipts AND creates a lots master row if Expiry_Da...	

6.9.3 Receipt History (last 50)

Below the form, an AgGrid showing recent receipts for this site: rowid - Date - SAP - Material - Quantity - Supplier - PR_Number - Expiry_Date.

6.10 HOD Portal -> [!] Shelf-Life

Lots at risk of expiry.

6.10.1 3-card stat strip

Card	Source
Expired	Lots with days_left < 0
? Critical (?30d)	Days_left 0-30
Warning (?90d)	Days_left 31-90

6.10.2 Action-required banner

When expired_n > 0: red banner instructing physical isolation + disposal.

6.10.3 Shelf-life table (top 30, sorted by days-left ascending)

Columns: SAP - Material - Lot - Qty - Expiry (color-coded) - Days Left - Status pill

6.10.4 ? Log Disposal button

Bulk-logs an audit event for the expired count. Note: this does NOT post a stock adjustment automatically - for proper inventory reduction, use the Adjustments flow with reason expired_disposal.

6.11 HOD Portal -> Notifications

Manual WhatsApp sends + alert threshold tuning.

6.11.1 Send Manual WhatsApp card

Field	Purpose
Recipient phone number	E.g., +966 5X XXX XXXX
Message textarea	Up to several hundred characters
Send WhatsApp button	Queues to whatsapp_queue with status='pending'; the background worker picks it up. Audit-logs MANUAL_WHATSAPP.

6.11.2 Alert Thresholds card

3 sliders writing to app_settings:

Slider	Range	Default	Stored key
Low stock alert (days of supply)	1-60	5	low_stock_days
Burn-rate warning (days remaining)	1-60	7	burn_alert_days
Expiry warning (days before)	1-120	30	expiry_warn_days

Save Thresholds button - persists + audit-logs UPDATE_THRESHOLDS.

> Note: The Notification Log table moved to Admin Portal -> WhatsApp Console for global visibility.

6.12 HOD Portal -> [OK] My Requests

Outbound cross-site requests YOU have made.

6.12.1 Outbound requests table (styled HTML)

Columns: # - To Site - SAP - Qty - Status pill - Created

6.12.2 Mark Incoming Transfers as Received

When at least one request has status='approved':

Element	Purpose
Select Approved Request selectbox	List of approved request IDs
Confirm Delivery Received button	Calls update_request_status(status='fulfilled') + busts caches. This is where you confirm the transfer physically arrived.

6.15 HOD Portal -> ? DN Approvals (NEW in v3.0)

This is your approval queue for Delivery Notes inbound to your site. The Warehouse has prepared them, Logistics has approved the delivery date - now you confirm the content (what's actually arriving, in what qty).

6.15.1 Empty state

If nothing's pending: an empty-state card reads **"No DNs awaiting your approval - They appear here after Logistics signs off the delivery date."** No action needed.

6.15.2 Per-DN cards

Each pending DN renders as its own bordered container with two columns:

Left column:

- Header line: DN <number> - PO <number> - From Warehouse <id>
- Subline: DN Date - Vehicle No - Driver Name
- View lines expander: Read-only preview of every line item (Material_Code, Description, Qty, UOM, rl_bl_family, Lot_Number, Expiry_Date, Remarks)

Right column (Decide popover):

- Notes textbox (required if rejecting)
- [OK] Approve button (primary) - flips DN to pending_sk and mirrors lines into pending_receipts so SK sees them
- [X] Reject button - flips DN to rejected with your reason; Warehouse gets pinged to redo

6.15.3 When to approve

- DN qty matches what your site needs (matches the PR line you originally raised)
- RL/BL family is correct
- Lot + expiry are acceptable (not over-expiry)
- The originating PO is the right one (not a misroute)

6.15.4 When to reject

- Wrong qty (e.g. site asked for 100, DN says 200 - too much exposure)
- Wrong material (the Material_Code doesn't match the PR you raised)
- Lot has insufficient remaining shelf life
- Vehicle / driver details suggest a routing problem

Always include a clear rejection reason. The Warehouse user sees the reason verbatim in their bell inbox and on the bounced DN. Sloppy reasons cost a rebuild cycle.

6.16 HOD Portal -> ? In-Transit (NEW in v3.0)

A read-only window onto the procurement chain for your site. Use this tab when a user asks **"when is X arriving?"** or **"why didn't Y arrive yesterday?"**. Three sub-tabs.

6.16.1 Sub-tab: ? Active in-transit

KPI strip at the top with counts per pipeline state:

- At Logistics - DN drafted by Warehouse, waiting on Logistics date approval
- Logistics approved - date confirmed, waiting on YOU
- Awaiting my approval - same as Logistics approved (highlighted gold)
- Pending SK receipt - you've approved, SK has it in their tab

Below the KPI strip, each in-transit DN renders as a card:

Element	Purpose
DN header	DN <number> (gold) + RL/BL chip (orange for RL, purple for BL)
Subline	PO <number> - Warehouse <id> - ETA <date> - <line count> line(s), <total qty> units
Status pill (right)	Colour-coded pipeline state
View lines expander	Read-only line preview
? Request reschedule popover	Date picker (defaults to ETA + 3 days, min=today) + reason textarea + Submit button. Submits to Logistics.

The reschedule UI is deliberately frictionless:

- Date is pre-filled so a quick "+3 days" submission is one click
- min_value=today prevents accidentally picking yesterday
- Same-date submission warns instead of wasting a round-trip
- Caption " Goes to Logistics" makes the destination unambiguous

6.16.2 Sub-tab: ? My reschedule requests

KPI strip: Pending / Approved / Rejected counts.

Custom table showing your full reschedule history for THIS site:

Column	Source
PO No.	the linked PO
DN No.	the linked DN (if any)
From	current_date when you raised the request
Requested	the new date you asked for (gold, monospace, with arrow)
Reason	your justification (tooltip-truncated)
Status	pill - pending / approved / rejected
Decided by	Logistics user who handled it
Notes	decision notes from Logistics

6.16.3 Sub-tab: ? Force-closures affecting me

Read-only audit table showing every PR / PO / line that Logistics force-closed, scoped to your site. Three-way fallback join means closures on records with NULL Site_ID still appear correctly via their PR

or PO parent.

Column	Source
Type	"PR closed" / "PO closed" / "Line closed" badge
Target	The closed ref (PR number / PO number / line id)
PR No.	Linked PR (if applicable)
PO No.	Linked PO (if applicable)
Reason	Logistics' force-close reason
Closed by	Logistics user who closed it
When	Timestamp

50 most recent shown.

6.13 HOD - Use Cases

Use Case 1: Approve and commit the day's consumption

1. HOD Portal -> EOD Commit
2. Review the 4-card stat strip
3. Sort/filter using the filter pills
4. For each row: read the full detail card (line 1 = essentials, line 2 = all fields)
5. Click [OK] to approve or [X] to reject individual rows
6. Optionally click [OK] Approve All Pending for a clean batch
7. Click Commit EOD to Master
8. If pre-flight blocks: read the violation table, close, fix the offending rows (reduce qty, drop, or receive stock first), then re-open
9. If pre-flight passes: type COMMIT exactly, click Confirm Commit
10. Watch the ? balloons.

Use Case 2: Approve a Store Keeper's physical count adjustment

1. HOD Portal -> Adjustments
2. Read the pending table - variance and reason are color-coded
3. For each adjustment, decide: did the Store Keeper count accurately?
4. [OK] Approve -> system posts a synthetic ledger row (receipt or consumption), live stock updates immediately
5. [X] Reject -> audit-logged; Store Keeper sees a status update

Use Case 3: Request material from another site

1. HOD Portal -> Cross-Site
2. Pick target site -> pick material -> see live availability
3. Add to cart, repeat
4. If cart > 5 items: an automatic WhatsApp escalation to the target HOD will fire on submit. Document why in Justification.

5.

Click Submit All Requests to Admin

6. Watch the [OK] My Requests tab - when status flips to "approved", you can mark it as fulfilled once received

Use Case 4: Approve an incoming bulk request from another HOD

1. HOD Portal -> Cross-Site -> scroll to Incoming Cross-Site Requests
2. Review the request table
3. [OK] Approve -> physical handoff begins
4. [X] Reject -> use when the request is unjustified or your stock is too tight

Use Case 5: Raise a Purchase Request

Option A: manual entry

1. HOD Portal -> Purchase Requests
2. ? Create New PR
3. Type PR Number, pick Material (auto-fills SAP, MatCode, UOM), Qty, Supplier, Est. Cost SAR, Notes
4. Click Create PR Draft

Option B: PDF upload

1. ? Upload PR PDF
2. Drop the PDF in
3. Click Process Upload
4. System auto-extracts PR number + items, creates pr_master rows

Use Case 6: Notify logistics about a pending PR

1. HOD Portal -> Purchase Requests
2. Select open PR in the email dropdown
3. ? Draft Outlook Email -> Mail/Outlook opens with HTML table pre-filled
4. Review/send
5. Also: Download PR PDF for your records

6.14 HOD - FAQ

Q: The EOD commit modal won't let me commit - keeps showing the violation table. A: Pre-flight is blocking because consuming the staged amount would create negative stock. Either: (a) reduce qty on the violating rows in the EOD table, (b) commit pending receipts first to top up stock, (c) raise a stock adjustment to correct any system-vs-physical discrepancy.

Q: A Store Keeper claims they submitted but I don't see anything. A: Check the filter pills (you may be filtering by Approved/Rejected only). Switch to "All" or "Pending".

Q: I approved an adjustment but the live stock didn't change. A: It should change within 30 seconds (cache TTL). Refresh the page. If it still doesn't, check audit log for the APPROVE_ADJUSTMENT entry to confirm the synthetic ledger row was posted.

Q: I got a WhatsApp about a "bulk cross-site request" - what is it? A: Another HOD at a different site has requested more than 5 items from your site. Go to Cross-Site -> Incoming to review.

Q: My burn rate forecast looks wrong - items aren't appearing. A: Items with fewer than ~3 consumption events in the last 30 days have insufficient data and may be omitted. Increase the window via Settings if needed.

Q: I can't see a receipt I just approved. Did it work? A: Approving via "Approve All" commits all pending receipts at once. Check the Live Dashboard for updated stock. The receipt now lives in receipts table, not pending_receipts.

Q: The Outlook email button does nothing on my Mac. A: Make sure Mail.app or Outlook is installed and the system has permission to open it. The fallback is mailto: (plain-text monospace table) which works in any browser default mail handler.

Q: PR PDF upload didn't recognize some items. A: The extractor matches material codes in the format GI-XXXXXXX. Items not in your inventory's Material_Code field will be flagged in the warning. Add the missing items to inventory first.

Q: Cross-site request stuck in 'pending' - why isn't Admin approving? A: Admin sees all pending in Admin Portal -> Pending Requests. If urgent, queue a WhatsApp from Notifications. Or for >5 items, the target-site HOD can also approve directly.

Q: I want to see all FEFO overrides logged by my Store Keepers. A: Filter Admin -> ? Audit Logs by action FEFO_OVERRIDE. Or query the consumption table for FEFO_Override IS NOT NULL.

7. Admin Manual

The Admin is the system owner. The Admin Portal has 11 tabs as of v3.0 - the original 10 (Overview, Pending Requests, Global Sites, Users, Master DB Editor, Audit Logs, WhatsApp Console, Settings, Access Control, Reports & Bugs) plus the new ? Logistics Oversight tab. Admins do NOT see HOD Portal (intentional - Admin uses Admin Portal for cross-site work). Admins CAN see the Logistics Portal and Warehouse Portal as shadow access - when entering the Warehouse Portal, a sidebar dropdown lets the admin pick which warehouse to view as.

7.1 Pages visible

- Live Dashboard (global - sees all sites combined)
- Entry Log (can stage entries themselves)
- Admin Portal - detailed below
- Reports (with "All Sites" filter unlocked)

7.2 Admin Portal overview

7.2.1 Page header

render_brand_header_admin(...) - gold subtitle + pulse status chip (green "All systems operational" or amber "Degraded - see Overview" if global low-stock count > 20).

7.2.2 Hero strip (3 cards)

Card	Source
Sites managed	Count of distinct sites
Pending cross-site requests	Count from requests where status='pending'
Critical items (all sites)	low + expiry count summed globally

7.2.3 Tab strip (9 tabs)

1. ?? Overview
2. Pending Requests
3. ? Global Sites
4. ? Users
5. ?? Master DB Editor
6. ? Audit Logs
7. WhatsApp Console
8. Settings
9. ? Access Control

7.3 Admin Portal -> ?? Overview

System health at a glance.

7.3.1 4-card KPI strip (technical)

Card	Source
?? DB size	File size of gi_database.db on disk (MB)
? Users	Count of non-suspended users
Total transactions	consumption + receipts row count
? Audit events	All-time count from system_audit_log

7.3.2 4-card valuation strip (financial)

Card	Source
Total stock value	cached_total_inventory_value() (all sites)
? Biggest-value site	cached_value_by_site().iloc[0] + share of total
30-day consumption value	cached_consumption_value(days=30)
Pending receipts value	Placeholder showing source: pr_master.Est_Cost_SAR

7.3.3 Service Health card

Per-service row with a pulse dot + status + note:

Service	Up indicator	Source
SQLite Database	Always up if reached	WAL mode - DB size MB
WhatsApp Queue	If whatsapp_queue reachable	Pending count or "queue clear"
Ollama / AI	OLLAMA_AVAILABLE if	"ready" / "not reachable"
Mail / SMTP	MAIL_ENABLED	"Outlook + mailto fallback"

7.3.4 Database Stats card

8 row counts:

- Inventory items
- Consumption rows
- Receipt rows
- Pending issues
- Pending receipts
- Open PR lines
- WhatsApp queue size
- Audit events

7.3.5 Live Activity Feed (last 12)

Per-row card from system_audit_log:

- // severity icon (derived from action_type via _severity_from_action())

—

Timestamp (gold monospace)

- Action name
- Target table pill
- "<user> - <details>" line

7.4 Admin Portal -> Pending Requests

Approve or reject cross-site material requests from HODs.

7.4.1 Pending requests data editor

Editable table with a ?? Select checkbox column (all other columns disabled). Source: `get_pending_requests(status='pending')`.

7.4.2 Admin Notes text input

Optional for approve - required for reject. Free text passed to `update_request_status()`.

7.4.3 Action buttons

Button	Action
[OK] Approve Selected	For each selected row: looks up Material Code/Description, updates <code>requests.status='approved'</code> . Then for each unique requester: queues a WhatsApp with item list + admin notes. For each unique targ...
[X] Reject Selected	Same multi-row loop but sets <code>status='rejected'</code> . Notes are MANDATORY for rejection. WhatsApp queued to each rejected requester with reason.

7.5 Admin Portal -> ? Global Sites

Cross-site inventory viewer (read-only).

Element	Purpose
Site selectbox	"All Sites (Global)" or pick one
Inventory dataframe	Calls <code>cached_live_inventory(site_id=...)</code>

Useful before approving a cross-site request to confirm the target site really has the stock.

7.6 Admin Portal -> ? Users

Delegates entirely to `auth.render_user_management_tab()` which provides:

Section	Purpose
Pending registration requests	Approve / reject self-registered users
Active users table	All users with role, site, phone
Edit user form	Change role, site, phone, suspend/activate
Add user form	Create user directly (Admin shortcut)
Password reset action	Generates temporary password, audit-logged

7.7 Admin Portal -> ?? Master DB Editor

This is the most powerful tab. Use with care. Lets Admin view, edit, and modify any table's data and schema.

7.7.1 Table selector

Element	Purpose
Select Table dropdown	Lists every user table in the SQLite DB. Includes new ones: lots, stock_adjustments, bug_reports, app_settings, etc.

7.7.2 Action radio

3 modes:

Mode A: View / Edit Data

Element	Purpose
Row count caption	"<N> rows in <table>"
? Export as PDF button	Generates generate_universal_pdf() of the whole table
Data editor table	Edit any cell. Password hashes are masked as *****.
Print Label checkbox column	Only on inventory table - select items to print QR labels
Save Table Updates button	DELETE the table, INSERT_ALL from the edited dataframe. Audit-logs DB_EDIT. Busts inventory + settings caches.
?? Generate QR Labels for Selected button	Only on inventory - generate_qr_labels_pdf() for items where label checkbox is checked
Download QR Labels PDF button	Appears after generation

> [!] The Save action is destructive. It deletes all rows and re-inserts. If anything fails mid-write you may lose data. Industry best-practice is to make ledger tables immutable; this tab bypasses that - Admin discretion required.

Mode B: ? Add New Entry

Form generation depends on the table:

—

For users: warns to use User Management tab instead (safer)

- For receipts: opens the full Logistics Receipt form (Site, Open PR linker, Material, Qty, Date, Expiry, Supplier, Remarks). Posts via process_receipt_delivery -> also creates lot if Expiry set.
- For any "transaction table" (has SAP_Code, isn't inventory): 2-section form - Section 1 picks material (shows MatCode + UOM info card), Section 2 dynamically generates inputs for every editable column.
- For inventory itself: dynamic 3-column form for all editable columns.

Mode C: Manage Columns

Element	Purpose
? Add Column section	Column name text input + Add button -> ALTER TABLE ADD COLUMN <name> TEXT
?? Rename Column section	Column dropdown + new name + Rename button -> ALTER TABLE RENAME COLUMN
? Drop Column section	Column dropdown + Delete button -> ALTER TABLE DROP COLUMN (SQLite supports this only on recent versions)

> The users table is protected - schema changes are blocked here ("managed by auth.py").

7.8 Admin Portal -> ? Audit Logs

Complete forensic record.

7.8.1 Filter row

Filter	Source
User selectbox	DISTINCT usernames + "All Users"
Action selectbox	DISTINCT action_types + "All Actions"
Target selectbox	DISTINCT target_tables + "All Targets"
Limit selectbox	50 / 100 / 500 / 1000

7.8.2 Search text input

Substring match across details + username + action_type.

7.8.3 Audit table (styled HTML)

Columns: // severity - Timestamp - User - Action (color-coded by severity) - Target pill - Detail (truncated, tooltipped)

Severity heuristic (from action_type):

- Critical (): any action containing FAIL, REJECT, DELETE, PURGE, EMERG, ROLLBACK, RESET, DESTRUCTIVE
- Warning (): SUSPEND, REVOKE, ROTATE, FLAG, DOWNGRADE, WARNING

—

Info (): everything else

Expired-row red tint on critical events.

7.9 Admin Portal -> WhatsApp Console

Outbound message queue + manual sends + thresholds + event mapping.

7.9.1 Queue stats strip (4 cards)

Card	Source
[OK] Sent	whatsapp_queue where status='sent'
? Pending	status='pending'
Processing	status='processing'
[X] Failed	status='failed'

7.9.2 Send Manual WhatsApp card

Field	Purpose
Recipient phone	E.g., +966 5X XXX XXXX
Message textarea	The message to send
Send WhatsApp button	Queues to whatsapp_queue + audit-logs MANUAL_WHATSAPP

7.9.3 Alert Thresholds card (global)

Same 3 sliders as HOD Notifications (low_stock_days / burn_alert_days / expiry_warn_days). Persists to app_settings. Admin's value is the global default.

7.9.4 Event -> Recipient (current wiring) - read-only summary

A reference table showing what auto-triggers exist in the codebase:

Event	Recipient	Role
Issue staging submitted	Site HOD	hod
Pending receipt submitted	Site HOD	hod
EOD committed	Site HOD	hod
Cross-site request created	All admins	admin
Cross-site bulk (>5 items)	Target site HOD	hod
Cross-site request approved	Requesting HOD	hod
Cross-site request rejected	Requesting HOD	hod
Returnable item overdue	Store Keeper	store_keeper
New access request	All admins	admin
Access request approved	Requesting user	store_keeper
Post-EOD low stock alert	Site HOD	hod

7.9.5 Outbound Queue Log (last 80)

Styled HTML table from `get_whatsapp_log(limit=80)`: Status pill - Recipient (monospace) - Message (truncated with tooltip) - Queued at - Sent at

7.10 Admin Portal -> Settings

7.10.1 Dropdown Manager - Work Types (expander, expanded by default)

Element	Purpose
Current Work Types caption	Comma-separated list from <code>system_settings</code>
New Work Type Name text input	Add a new option
Add to Dropdown button	INSERT into <code>system_settings</code> (category='Work_Type') + busts settings cache

7.10.2 Maintenance Mode card

Element	Purpose
Enable maintenance mode toggle	Persists to <code>app_settings.maintenance_mode='1'</code>
Status caption	"ACTIVE - Non-admin sessions will be told to come back later" or "Off"

Audit-logs TOGGLE_MAINTENANCE. (Enforcement at login is in `auth.py` and `main.py`.)

7.10.3 ?? Database Backup card

Element	Purpose
Last manual backup caption	From <code>app_settings.last_backup_at</code>
Backup Now button	Calls <code>shutil.copy2()</code> of <code>gi_database.db</code> into <code>backups/gi_database_YYYYMMDD_HHMMSS.db</code> . Updates <code>last_backup_at</code> + audit DB_BACKUP.

7.10.4 ? Site Management

Element	Purpose
Sites table (styled HTML)	Site Name - Code (first 4 chars upper) - Users count - Status (always "Active")
? Add New Site expander	Type name + button -> INSERT into <code>system_settings</code> (category='Site') + audit ADD_SITE

7.10.5 [!] Danger Zone (red bordered card)

Element	Purpose
Purge old draft pending_issues card	"Delete all pending_issues rows older than 30 days that are still status='draft'"
Type PURGE to confirm	Text input must equal PURGE to enable
Run Purge button	Executes the DELETE, audit-logs PURGE_DRAFTS

7.11 Admin Portal -> ? Access Control

7.11.1 ?? Recent Sign-Ins (last 10)

Per-row card from system_audit_log filtered to LOGIN/LOGIN_SUCCESS/LOGIN_FAILED/LOGOUT:

- Pulse dot (green=success, red=failed)
- Username
- Action label
- Timestamp + details

7.11.2 ? Force Password Reset card

Field	Purpose
Target user selectbox	From users table
New password	Type input (password mask)
Confirm	Must match
Amber warning	"User must log in again immediately."
? Reset Password button	Validates length ? 8 + match -> hash_password(new_pwd) -> UPDATE users SET password_hash + audit FORCE_PASSWORD_RESET

7.11.3 Security Policy

Read-only 2-column grid showing:

- Auth backend (bcrypt cost=12)
- Session storage (Streamlit session_state in-memory)
- RBAC hierarchy (store_keeper < supervisor < hod < admin)
- WAL mode + busy_timeout=5000ms
- Password min length (8 characters)
- Audit retention (indefinite, manual purge only)

7.11a Admin Portal -> ? Logistics Oversight (NEW in v3.0)

Cross-site, read-only window onto the entire procurement chain. For actions, jump to the Logistics Portal (shadow access) - this tab is observation-only by design.

7.11a.1 KPI strip

Six cards at the top:

Card	Source
OPEN PRs	Count from list_prs_for_logistics() - awaiting PO issuance

Card	Source
OPEN POs	Count from list_pos(open_only=True)
ACTIVE DNSs	Count of DNSs in pipeline states (pending_logistics, logistics_approved, pending_hod, pending_sk)
VENDOR RETURNS	Open returns from list_vendor_returns(open_only=True)
RESCHEDULES	Pending reschedule decisions
FORCE-CLOSURES	Lifetime audit count

7.11a.2 Filters

Element	Purpose
Site dropdown	"All sites" or pick one - narrows every sub-tab
Warehouse dropdown	"All warehouses" or pick one - narrows DN view

7.11a.3 Six sub-tabs

Sub-tab	What's shown
PRs	Every active PR in the Logistics queue, filterable by site
POs	Every open PO with vendor, dates, total, status, source (manual/PDF)
? DNSs	Every active DN with warehouse, site, status, family
<-? Vendor Returns	Open returns awaiting vendor acknowledgement
? Force-Closures	100 most recent force-closure records with reason + closed-by
? Reschedules	Pending reschedule decisions Logistics hasn't acted on

7.11a.4 What it's NOT

- This tab cannot create / approve / reject anything. All mutation happens in role-specific portals.
- Admins who need to ACT (e.g. approve a reschedule because Logistics is on leave) should switch to the Logistics Portal - admin has shadow access there.

7.12 Admin - Use Cases

Use Case 1: Set initial Unit_Costs for valuation reports

1. Admin Portal -> ?? Master DB Editor
2. Select Table: inventory
3. Action: View / Edit Data
4. In the data editor, find Unit_Cost column (rightmost)
5. Type SAR cost for each item
6. Click Save Table Updates

7.

Wait ~60 sec for caches to refresh, then check Live Dashboard hero card "Total stock value"

Use Case 2: Approve a self-registered new user

1. Admin Portal -> ? Users
2. Pending requests section -> review the request
3. Edit role/site if needed
4. Click [OK] Approve -> user can now log in. WhatsApp alert auto-fires to the requester.

Use Case 3: Backup the database before a risky operation

1. Admin Portal -> Settings
2. Backup Now -> confirms file path: backups/gi_database_YYYYMMDD_HHMMSS.db
3. Note the timestamp in "Last manual backup" field
4. Now perform the risky operation (bulk import, schema change, etc.)
5. If something breaks: restore by copying the backup back over gi_database.db (with the app stopped)

Use Case 4: Investigate a suspicious series of FEFO overrides

1. Admin Portal -> ? Audit Logs
2. Action filter: FEFO_OVERRIDE
3. Search: e.g., a specific username
4. Review the detail column - see SAP, site, chosen lot vs FEFO-suggested, reason
5. If a pattern of unjustified overrides emerges, talk to the Store Keeper / HOD

Use Case 5: Configure WhatsApp alert thresholds

1. Admin Portal -> WhatsApp Console
2. Alert Thresholds card
3. Adjust sliders: low_stock_days, burn_alert_days, expiry_warn_days
4. Save Thresholds -> audit-logged. HOD threshold sliders read the same values.

Use Case 6: Approve a batch of cross-site transfers

1. Admin Portal -> Pending Requests
2. Review the editor table - sort by created_at, target_site, or item
3. Tick checkboxes for the rows you want to approve as a batch
4. Type Admin Notes (e.g., "Per Tuesday meeting, expedite to Site B")
5. [OK] Approve Selected -> triggers two WhatsApp dispatches per unique requester + target HOD
6. Watch the Notification log in WhatsApp Console to confirm sends

Use Case 7: Drop a deprecated column from a custom table

1. Admin Portal -> ?? Master DB Editor
2. Select the table
3. Action: Manage Columns
4. ? Drop Column section -> pick the column -> Delete Column

Use Case 8: Generate QR labels for new inventory items

1. Master DB Editor -> inventory
2. View / Edit Data mode
3. In the leftmost "Print Label" column, tick items
4. Scroll to ?? QR Code Label Generator section below
5. Click ?? Generate QR Labels for Selected
6. Download QR Labels PDF
7. Print on label paper, stick to shelf bins -> store keepers can now scan them on mobile

7.13 Admin - FAQ

Q: How do I add a new site? A: Admin Portal -> Settings -> ? Site Management -> ? Add New Site. New sites are immediately available for user assignment.

Q: How do I rotate a user's password without their input? A: Admin Portal -> ? Access Control -> ? Force Password Reset. Pick user, type new password twice, click reset. They'll need to log in again with the new credentials.

Q: Audit log filter has too many users - how do I find a specific user's activity? A: Use the search text input at the bottom of the audit filter row - searches across user + action + details.

Q: WhatsApp queue is stuck - messages aren't being sent. A: The background worker (whatsapp_worker.py) must be running separately (not via Streamlit). Check via terminal: python whatsapp_worker.py. The queue table will fill but won't drain otherwise.

Q: My DB Editor save crashed mid-way and lost rows. A: Restore from backups/. The Save action does DELETE-then-INSERT and is not crash-safe. Always backup before bulk edits.

Q: Maintenance Mode is on but non-admins can still log in. A: Enforcement requires a check in auth.py. The toggle persists; ensure the auth gate respects app_settings.maintenance_mode.

Q: Can I delete an audit log entry? A: No - they're meant to be immutable. If you need to free space, do a SQL purge with a date filter via DB Editor -> Manage Columns... but this is itself audited.

Q: A cross-site request is stuck pending and the requester is asking why. A: Admin Portal -> Pending Requests. Confirm it's actually in the queue. If yes, approve or reject with notes. The HOD will get a WhatsApp.

Q: I want to set per-user 2FA. A: Currently unsupported in the security model (placeholder shown in Access Control). Future enhancement.

Q: How do I know if Ollama (local AI) is up? A: Admin Portal -> ?? Overview -> Service Health card. If "Ollama / AI" pulses green, it's reachable.

Q: What's the difference between approving a Pending Receipt vs Pending Issue? A: Receipts ADD to stock (your team RECEIVED material). Issues SUBTRACT from stock (your team CONSUMED material). Both flow through approval. Issues commit via EOD (batch); Receipts commit individually via Pending Receipts tab.

Q: Stock_Value column shows nothing on the dashboard. A: Inventory items have Unit_Cost = 0. Use

DB Editor to set costs.

Q: A HOD set thresholds different from mine - whose wins? A: Both HOD and Admin write to the same `app_settings` keys. Last-writer-wins. Decide a policy and lock it via convention.

Q: How big can the database grow before performance degrades? A: SQLite with WAL comfortably handles 10-25 concurrent users. The codebase is structured so `database.py` is the only SQL surface - migrating to Postgres later is a contained change.

8. Reports Module - Detailed Reference

Available to: Supervisor, HOD, Admin. Site scope: locked for Supervisor + HOD; "All Sites" available to Admin.

8.1 Page header + 4 tabs

render_brand_header("Reports & Analytics"). Tabs:

1. Generate Report
2. ? Scheduled
3. AI Insights
4. ? Archive

8.2 Reports -> Generate Report

8.2.1 Date + filter row

Field	Purpose
From date	Start of date window
To date	End of date window
Site filter	Locked to own site for non-Admin; "All Sites" + dropdown for Admin
Format	PDF / Excel / CSV

8.2.2 Report type selector

A grid of 9 selectable cards:

Report	Description	Source
Daily Consumption	All material issues in the window, grouped by work type	report_daily_consumption()
? Monthly Summary	Per-SAP opening, issued, received, closing stock + SAR value columns (Issued, Received, Closing)	report_monthly_summary()
[!] Low Stock Alert	Materials below minimum with shortfall	get_low_stock_items()
Burn Rate Analysis	30-day trends, days-of-supply per item	cached_burn_rate_and_forecast()
Inventory Valuation	Per-item Stock_Value sorted descending; summary with top-10 share	get_inventory_valuation()
Shelf-Life / Expiry	Lots by expiry -	

expired/critical/warning buckets

get_short_dated_stock()

Report	Description	Source
PR Status Report	All PRs with status, workflow_state, Est_Cost_SAR	report_pr_status()
[OK] FEFO Compliance	Audit of FEFO adherence - picks vs oldest lot	report_fefo_compliance()
? Full Audit Report	Complete system_audit_log for the date range	report_audit_export()

8.2.3 ? Generate Report button

Calls `_run_report(type, from, to, site_id)` -> returns (DataFrame, summary_dict). Loading state shows shimmer.

8.2.4 Preview section (after generation)

Element	Purpose
Report title + subtitle	"<Type> - <site>" + "<from> -> <to>"
v Download <fmt> button	Encodes via <code>generate_report_pdf/excel/csv</code> and triggers <code>st.download_button</code>
Summary cards	Auto-generated from the summary dict keys
Bar chart	For Burn Rate and Daily reports
Preview table	Styled rows with status-color cells

8.2.5 ? Email Delivery section

Field	Purpose
Recipients text input	Comma-separated emails
? Send Email button	Generates and emails the report
WhatsApp button	Sends a brief summary via WhatsApp

8.3 Reports -> ? Scheduled

Manage automated recurring reports.

8.3.1 + New Schedule button (expands form)

Field	Purpose
Report Type	One of the 9
Frequency	Daily 06:00 / Daily 17:00 / Weekly Mon 07:00 / Monthly 1st 06:00
Format	PDF / Excel / CSV
Recipients	Comma-separated users

8.3.2 Schedule cards

Each schedule renders as a card with:

—

Report icon + label

- Frequency - recipients
- Last run timestamp
- Active/Paused pill + toggle
- ? Run Now button (manual trigger)
- ? Delete button

8.4 Reports -> AI Insights

If AI_ENABLED and Ollama is running with llama3.1:8b:

8.4.1 Header bar

+ "AI-Powered Inventory Analysis" + BETA pill + intro text. Shows "Analysing data..." shimmer during regen.

Button	Action
Regenerate	Re-runs all 5 fixed SQL probes + LLM commentary

8.4.2 5 Insight cards (collapsible)

Each card has a left-border in severity color, then:

Element	Purpose
Icon + Title	"Abnormal Consumption Spike - MAT-XXXX"
Severity pill (Critical/Warning/Positive)	Color-coded
Confidence % with progress bar	LLM-reported confidence
Right-side metric callout	Headline number with sub-label
Body paragraph	LLM-generated explanation grounded in the SQL probe result
? Recommendations list	Up to 3 numbered actions
? Share button	Shares with HOD team
[OK] Add to Actions button	Logs as a follow-up

The 5 fixed probes:

1. Abnormal consumption spike (vs trailing average)
2. Items approaching reorder
3. FEFO compliance rate
4. Procurement cost optimization (supplier consolidation)
5. Inventory health score

If Ollama is not running: a graceful fallback explains setup steps.

8.5 Reports -> ? Archive

Permanent record of previously generated reports.

8.5.1 Search row

Element	Purpose
Search archive text input	Substring match on name / type
Total counter caption	"N reports - X KB total"

8.5.2 Archive table

Columns: Report Name - Type pill (colored by report type) - Generated date - By - Format icon + label - Size - Actions

Per-row Actions:

- v Download
- ? Re-email
- ? Delete (audit-logged)

9. Automated Notifications - WhatsApp & Email & In-app Bell

The system automatically queues messages on key events. There are now THREE notification surfaces:

1. WhatsApp queue - whatsapp_queue table, drained by whatsapp_worker.py. Gated by config.WHATSAPP_ENABLED master switch + config.WHATSAPP_TRIGGERS per-event dict. Flip a key to False to silence WhatsApp for that event without touching in-app behaviour.
2. In-app notifications bell (NEW in v3.0) - app_notifications table, surfaced via the sidebar bell described in §3.6. ALWAYS fires regardless of WhatsApp toggle.
3. Email - same Outlook / Mail.app / mailto: + SMTP paths as before.

9.1 WhatsApp triggers (auto-fire)

Event	Recipient	Trigger location	Notes
Store Keeper submits issue batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits receipt batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits stock count	Site HOD	Entry Log Stock Count Submit	Includes variance + reason
Store Keeper performs FEFO override	Site HOD	Entry Log Add to Grid	Includes chosen vs FEFO-suggested lot +
HOD commits EOD	Site HOD (self)	post-commit	Only if low-stock items result
HOD cross-site request created	All Admins	Submit cart	Includes target site + count
HOD cross-site request, > 5 items	Target Site HOD	Submit cart	Bulk escalation
Admin approves cross-site requests (batch)	Requesting HOD	Approve Selected	Includes approved items + admin notes
Admin approves cross-site requests (batch)	Target Site HOD	Approve Selected	"TRANSFER ORDER - pack and ship"
Admin rejects cross-site requests (batch)	Requesting HOD	Reject Selected	Includes rejection reason
Returnable item overdue	Borrower + Store Keeper	Background scheduler	Time-driven, not in current scope
New self-registration request	All Admins	Registration form	Includes username + role + site
Self-registration approved	Requesting user	Admin User Mgmt	"ACCESS GRANTED - Welcome"
Manual send	Free	HOD Notifications / Admin WhatsApp Console	Audit MANUAL_WHATSAPP

Event	Recipient	Trigger location	Notes
PR submitted to Logistics (v3.0)	Logistics role	HOD PR tab -> ? Submit PR(s) to Logistics	Event key pr_submitted_to_logistics
PO issued (v3.0)	Site HOD	Logistics -> Save PO	Event key po_issued
PO assigned to Warehouse (v3.0)	Warehouse users at that WH	Logistics -> Assign	Event key po_assigned_to_warehouse
Warehouse acknowledged (v3.0)	Logistics	Warehouse -> [OK] Acknowledge	Event key warehouse_acknowledged (off by default)
Warehouse received goods (v3.0)	Logistics	Warehouse -> Record receipt	Event key warehouse_received
DN logistics approved (v3.0)	Site HOD	Logistics -> [OK] Approve DN	Event key dn_logistics_approved
DN HOD approved -> SK (v3.0)	Site SK	HOD -> ? DN Approvals -> [OK] Approve	Event key dn_auto_generated (reuses slot for SK ping)
DN received at site (v3.0)	Logistics + Warehouse	SK -> [OK] Mark as Received	Event key dn_received_by_sk (off by default)
Reschedule requested (v3.0)	Logistics	Warehouse / HOD -> ? Request reschedule	Event key reschedule_requested
Reschedule decided (v3.0)	Requester	Logistics -> [OK] Approve / [X] Reject	Event key reschedule_decided
Vendor return raised (v3.0)	Logistics	Any role -> <-? Raise return	Event key vendor_return_raised
PR force-closed (v3.0)	Admin + originating Site HOD	Logistics -> ? Force-Close	Event key pr_force_closed, severity critical
PO force-closed (v3.0)	Admin + originating Site HOD	Logistics -> ? Force-Close	Event key po_force_closed, severity critical
Delivery reminder T-2 / T-1 / T-0 (v3.0)	Logistics + HOD + Warehouse (per DN)	Daily sweep job - see §9.3	Event keys delivery_reminder_t_minus_2/_minus_1/_zero

9.2 Email triggers

Event	Trigger location	Mechanism
Logistics PR follow-up	HOD PR tab -> ? Draft Outlook Email	Mac: AppleScript-driven Mail.app with HTML table. Windows: COM Outlook with HTMLBody. Fallback: mailto: with monospace plain-text...
EOD report email	mailer.py SMTP	Configurable via .env (SMTP_SERVER, SMTP_USER, SMTP_PASSWORD)
Report delivery (scheduled)	Reports -> Scheduled	Same SMTP pipeline

9.3 In-app notifications bell (NEW in v3.0)

See §3.6 for the UI. Backed by the `app_notifications` table; queried with role + site + warehouse scoping. Per-event severity (info / warning / success / critical) drives the colour-coded left border on each card.

9.4 Delivery reminder daily sweep (NEW in v3.0)

A once-per-day job in `whatsapp_worker.run_worker_loop()` calls `sweep_delivery_reminders()` which fires T-2 / T-1 / T-0 reminders for upcoming deliveries:

Watched	When	Severity
<code>purchase_orders.Expected_Delivery</code>	T-2 / T-1 / T-0	warning / warning / critical
<code>delivery_notes.DN_Date</code>	T-2 / T-1 / T-0	warning / warning / critical

PO reminders ping: Logistics + originating Site HOD. DN reminders ping: Logistics + Site HOD + Warehouse user(s) at the receiving warehouse.

Idempotency - the sweep cannot double-fire for the same target on the same day. Two guards:

1. `UNIQUE(ref_type, ref_number, target_date, offset_days)` on the `delivery_reminders_sent` table.
2. Day-marker stored in `app_settings.delivery_reminders_last_run` - the 60-sec worker poll loop skips the sweep query entirely on second-and-later ticks of the same day.

Restarting the worker mid-day is safe. Re-running the sweep manually on the same day fires zero new notifications.

Customising the cadence - currently the offsets are hardcoded (2, 1, 0) in `database.sweep_delivery_reminders()`. A configurable offsets list is on the v3.0 backlog (see `handoff.md` §3 item 25).

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (PK), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override,	? subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current_Stock} = \text{Total_Received} - \text{Total_Consumed} + \text{Total_Returned}$ - computed live in v_site_stock view.

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	API tokens for offline PWA
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4a Procurement chain tables (NEW in v3.0)

Table	What it stores	State machine
warehouses	Master of receiving locations	active / inactive
vendors	Supplier master	active / inactive
purchase_orders	PO header (PO_Number UNIQUE)	open -> partially_delivered -> delivered -> closed / force_closed / cancelled
po_items	PO line items with rl_bl_family tag	open -> partially_delivered -> delivered / returned / closed / force_closed
po_shipment_schedule	Parsed PO Annexure rows	pending / shipped / delivered / delayed / cancelled
po_assignments	Logistics -> Warehouse routing	assigned -> acknowledged -> partial / received / closed / cancelled
delivery_notes	DN header (DN_Number UNIQUE)	DN state machine: draft -> pending_logistics -> pending_hod -> pending_sk -> received (or rejected from any pending)
dn_items	DN line items	pending / received / partial / returned / cancelled
po_returns	Vendor returns (raised by any role)	open -> vendor_acknowledged / resupplied / cancelled
po_reschedule_requests	Date-change asks	pending -> approved / rejected
po_force_closures	Force-closure audit log	(terminal - write-once)
app_notifications	In-app bell inbox	read_at IS NULL (unread) -> timestamp (read)
delivery_reminders_sent	T-2/T-1/T-0 idempotency log	(terminal - UNIQUE constraint)

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at site)
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for audit)
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	? Damaged / unusable	Physical damage
expired_disposal	? Expired - disposed	Expired and physically discarded
miscount_in	? Miscount - found extra	Found more than system shows
miscount_out	? Miscount - short	Found less than system shows
lost	? Lost / unaccounted	Genuinely missing
theft	? Suspected theft	Deliberate removal
return_to_supplier	<-? Returned to supplier	Quality issue, sent back
other	? Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit

in_progress - supplier engaged

- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	Substring RL-, RUBBER LINING, RUBBER-LINING in Material_Code OR Description
BL	Brick Lining	Substring BL-, BRICK LINING, BRICK-LINING, BRICK MATERIAL
NULL	Neither family	Default

Logic in `config.classify_rl_bl_family()`. NEVER a combo string - RL takes precedence if both tokens are present.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (? blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	Write-Ahead Logging (SQLite concurrency mode)
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)

Term	Meaning
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> ? Access Control -> ? Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13

13. 2026-06 Feature Update - What Changed

This section documents the upgrades shipped in the 2026-06 release. Everything above remains accurate except where this section overrides it.

13.1 Field-level changes

Change	Where	Behaviour
All form fields mandatory	Entry Log (Consumption / Receipt Staging), HOD Receive Material, Admin Add Entry forms	Every text/number/select input is required. Validation lists missing fields on submit.
Expiry Date is optional	SK Receipt Staging, HOD Receive Material	Marked (Optional). Leave blank for non-perishable items.
Remarks, Tank No., Serial No., PR Number	Same forms	No longer optional.

13.2 Live Dashboard column order

The Live Dashboard table now renders in this order (when columns exist):

SAP_Code -> Material_Code -> Equipment_Description -> UOM -> Opening_Stock -> Receipt -> Consumption -> Return -> Closing_Stock -> Minimum_Qty -> Unit_Cost -> Stock_Value -> Category -> Status

- Opening_Stock is now a configurable column on inventory. Default 0; admin can edit in DB Editor.
- Identity formula updated to $\text{Closing_Stock} = \text{Opening_Stock} + \text{Total_Received} - \text{Total_Consumed} - \text{Total_Returned}$.
- Material_Code now appears after SAP_Code in every report (Daily Consumption, Daily Receipts, Monthly Summary, PR Status, etc.) and in the HOD Pending Receipts approval list.

13.3 Entry Log access - Store Keeper only

The Entry Log page is now visible only to the store_keeper role. HODs review submissions in HOD Portal; Admins use Admin Portal. The page is hidden in the sidebar for other roles.

13.4 EOD Commit - checkbox confirmation

The "Confirm EOD Commit" dialog no longer requires typing COMMIT. Tick the confirmation checkbox and click Confirm Commit. Cancel still drops all pending state.

13.5 Material Category

Every inventory item now carries a Category. Categories: Consumable, Equipments, Utilities,

Maintenance, Others (default), Rubber materials, Tools, QC items.

- Admin Portal Add New Entry -> renders Category as a selectbox.
- Reports page -> Filter by Category dropdown alongside the SAR / cost-columns toggle. "All Categories" disables the filter.

13.6 Rubber MTC workflow

When a Store Keeper stages a receipt and the selected material's category is Rubber materials, the system shows:

- MTC Number text field (e.g. MTC-2026-001234)
- MTC Document file uploader (PDF / JPEG / JPG / XLSX)

Either field can be blank - the receipt still goes through. What changes is HOD-side visibility:

- HOD Portal -> Pending Receipts: a red banner lists rubber items received without an MTC.
- Click Draft Logistics Email to open a pre-filled email to the logistics team listing SAP, description, lot, qty.
- Sending (or clicking "Mark all as sent") flips the rubber rows to sent_to_logistics.

The logistics email recipient defaults to LOGISTICS_EMAIL env var (logistics@generalindustries.net if unset). On Windows it opens Outlook; on macOS it opens Mail.app; on Linux it opens the default mailto handler.

13.7 Document attachments - Entry Log + HOD DOC tab

Store Keepers can attach reference documents (PDF / JPEG / JPG / XLSX) on:

SK form	Doc number used	Notes
Consumption Log	Auto = DDMMYY of the date	Pick scope: "Whole entry (batch)" or "Specific date".
Receipt Staging	DN No. of the row (or manual override)	Falls back to DN-DDMMYY if no DN_No found.

Files are stored as BLOBs inside the database (authoritative copy) and mirrored to uploads/<Site_ID>/<doc_type>/<doc_number>/ for local browsing. The disk mirror is gitignored; only the DB BLOB is portable.

HOD Portal -> DOC is a new tab with three sub-tabs: Consumption / Receipt / <-? Return. Each shows period (From/To dates) and Doc Number text filters, with a per-file ?? download button.

The <-? Return sub-tab pulls from the new Return Items workflow (see §13.10), not from Returnable Items.

13.8 QR Label approval flow

The Admin DB Editor's QR generator (single-user, single-item) is unchanged. The new flow is two-step:

1.

Store Keeper -> Entry Log -> QR Label Request (new tab)

- Multi-select materials in one form.
- Per-item label quantity.
- Click Submit Batch for HOD Approval.

1. HOD Portal -> QR Approval (new tab)

- ? Pending sub-tab: select rows via checkbox, then [OK] Approve Selected or [X] Reject Selected.
- [OK] Approved sub-tab: Download QR Labels PDF for ALL approved generates one consolidated PDF.

13.9 Returnable Items - clarification

The Returnable Items tab is for temporary tool loans only (e.g. issuing a torque wrench to a worker who'll return it before EOD). It is **not** a way to return stock to the warehouse. No DN No., no document attachment.

For real returns (defective material going back to logistics), use the new Return Items tab - see §13.10.

13.10 Return Items workflow (NEW)

The new <-? Return Items tab (between Receipt Staging and Returnable Items in the Entry Log) handles real returns to the warehouse / logistics.

Store Keeper flow

1. The material picker is restricted to materials received in the last 30 days at the user's site.
2. If multiple receipts exist for the same SAP code, the SK is asked which receipt is being returned (Date / DN No. / Received Qty).
3. The system shows a locked summary of the original receipt: Date, DN No., PR, Lot, Received Qty.
4. The SK enters:
 - Return Quantity (capped at the original Received Qty)
 - Reason (work-types dropdown)
 - Return DN No.
 - Attachment - mandatory: the Return DN + any photos
9. To return material older than 30 days, tick "Override 30-day window" - the picker widens to 12 months and an Override Justification field appears. This routes to HOD as an explicit override request.
10. Submit -> request lands in HOD Portal <-? Returns tab. A WhatsApp ping goes to the site HOD if a phone number is on file.

HOD flow

1. HOD Portal -> <-? Returns lists every pending return with a card per row.
2. Rows that required an override are highlighted in red and show the SK's justification.
3. [OK] Approve -> writes a row to the returns ledger (so Current_Stock reduces by the returned qty, the dashboard Return column ticks up, and the entry shows up in monthly / consumption reports). Then

automatically opens the logistics email draft with item, qty, reason, and the original receipt's DN/PR/Lot context.

4. [X] Reject -> marks the request rejected. The SK sees this in their request history.

Dashboard / report impact

- The Return column on the Live Dashboard reflects approved returns (since returns is the source of truth).
- All existing reports (Daily Consumption, Daily Receipts, Monthly Summary, Audit) include returns via the same identity math.
- The HOD DOC tab <-? Return sub-tab lets the HOD browse all attached return documents.

13.11 Per-site Work Type and Tank No.

HOD Portal -> Site Config lets the site HOD add or delete Work Types and Tank Numbers scoped to their site. Empty per-site lists fall back to the global defaults.

13.12 WhatsApp worker - startup fix

The worker module no longer imports pywhatkit at module load (that pulled in heavy GUI deps and stalled local launch by tens of seconds). It now lazily imports pywhatkit only when an outbound message has no Twilio fallback. On Streamlit Cloud, Twilio handles delivery; on local desktop, pywhatkit is loaded the first time a message is queued.

If you see the spinner sitting on `_start_whatsapp_worker()` for more than ~3 seconds locally, check that you ran `pip install -r requirements.txt` after the update.

13.13 Category rename - Rubber Materials -> Surface Shields

The category that triggers the MTC-required workflow on Receipt Staging is now Surface Shields instead of "Rubber Materials". Everywhere the system used to look for Rubber materials, it now looks for Surface Shields. The behaviour is identical:

- SK selects a Surface Shields item on Receipt Staging -> MTC Number + MTC File uploader appear.
- Missing MTC = HOD sees the red banner on Pending Receipts with Draft Logistics Email.

The internal constant is now `MTC_REQUIRED_CATEGORY` in `config.py`. If you ever need to apply MTC enforcement to another category, change one string and the rest of the system follows.

13.14 WBS Master + WBS-aware Entry Log + WBS Report

- HOD Portal -> Site Config -> ? WBS Numbers lets the HOD add, close, or re-open WBS numbers for their site. Each WBS carries an optional Description and an active / closed status.
- Entry Log -> Consumption Log and Receipt Staging show a WBS Number dropdown filtered to the SK's site. If the HOD hasn't added any WBS yet, the SK sees a warning and a free-text fallback so work isn't blocked.

-

Reports -> ? WBS Report rolls everything up by WBS for a chosen date range, scoped to the user's site. Columns: WBS_Number, Consumption_Rows, Consumption_Qty, Consumption_Value_SAR, Receipt_Rows, Receipt_Qty, Receipt_Value_SAR. Sorted by consumption value descending.

- DB column wbs is now self-healed on consumption, receipts, pending_issues, pending_receipts. Old rows show as (no WBS) in the report; new rows carry the picked WBS through SK -> HOD commit untouched.

13.15 Site_ID badge in sidebar

Every signed-in user now sees their site code as a gold pill in the sidebar user card, next to the role label. This makes "wait, which site am I logged into?" impossible to get wrong, which matters when an Admin shadows a site for support.

13.16 Live Dashboard - live-typing filter on key columns

The dashboard filter row no longer waits for Enter. Typing into any of the four searchable columns - SAP_Code, Material_Code, Equipment_Description, Category - narrows the table on every keystroke (~180 ms debounce). The numeric columns deliberately have no filter input: text filters on numbers don't help and they made the page laggy.

Header reads "Filters (live) - searchable on SAP / Mat Code / Description / Category".

The package streamlit-keyup powers the live updates; if it's not installed, the dashboard falls back to plain st.text_input (still works, just needs Enter).

13.17 Sidebar Hub Assistant - role-aware Q&A

A new sidebar expander ? Ask Hub Assistant lets any signed-in user ask plain-English questions about the system. The answer is generated by the local Ollama llama3.1:8b model. Role filtering happens at the RAG context layer, not just the system prompt - a Store Keeper's question never sees the Admin chapter, so the model physically cannot answer about admin features even if asked.

Section visibility:

- Store Keeper: §1, §2, §3, §4, §11, §13
- Supervisor: SK list + §5, §8
- HOD: Supervisor list + §6, §9
- Admin: everything

The widget streams the answer token-by-token. Click Clear to reset.

13.18 GMT+3 (Asia/Riyadh) timestamps

Timestamps shown in the UI are now Riyadh local time. Background:

- New rows written through Python (datetime.datetime.now()) write Riyadh time directly because TZ=Asia/Riyadh is set in every host launchd plist.

-

Rows written through SQLite's DEFAULT CURRENT_TIMESTAMP (still UTC, per SQLite spec) are converted at display time via `config.utc_to_local()`.

- Affected views so far: Admin Pending Cross-Site Requests, Admin Audit Logs, Admin Live Activity Feed, Admin WhatsApp Console, HOD Cross-Site Incoming, HOD Pending Receipts. Other tabs still show UTC; tell the admin to file a ticket if any specific column matters and isn't converted yet.

Override the offset via the env var `GI_TZ_OFFSET_HOURS` if you ever roll out to a site outside KSA.

13.19 Self-host setup - `host\_setup/` folder

Path A (Mac + Cloudflare Tunnel) is now turnkey. The repo ships:

- `host_setup/launchd/*.plist.tmpl` - templates for the 4 background services
- `host_setup/scripts/install.sh` - one-shot install + load
- `host_setup/scripts/run_streamlit.sh` - wrapper that exec's Streamlit (avoids macOS exit-126 caused by direct caffeinate chain)
- `host_setup/scripts/restart_app.sh` - zero-downtime restart after git pull
- `host_setup/scripts/uninstall.sh` - unload + remove without touching data
- `host_setup/scripts/backup_db.sh` - SQLite online backup + iCloud Drive mirror + 14-day prune
- `host_setup/cloudflared_config.yml.example` - tunnel config template
- `host_setup/README.md` - full 45-minute setup playbook

See §14 below for the operations guide.

13.20 WhatsApp on macOS - Chrome via Accessibility-only AppleScript

The Mac worker now uses `open -a <browser> + osascript` with only System Events keystrokes (no tell application <browser>). This means:

- macOS asks for Accessibility permission once, on first send. Grant it. Done forever.
- No more "Python wants to control Google Chrome" Automation prompt on every message.
- Works with Chrome OR Safari - set `GI_WHATSAPP_BROWSER` env var. The worker plist defaults to chrome.

Two new env-controlled knobs:

- `GI_WHATSAPP_BROWSER` (default Google Chrome)
- `GI_WHATSAPP_WAIT_S` (default 15 - seconds between opening the URL and pressing Enter; raise on slow Macs)

When the standalone worker is running (`launchd com.gi.whatsapp-worker`), the embedded thread inside the Streamlit process is suppressed via `GI_SUPPRESS_EMBEDDED_WORKER=1` so the two don't race for the same queue rows.

13.21 v3.0 Procurement chain - what changed (2026-06 round 3)

The largest single feature batch since launch. Fully additive - no edits to existing SK / HOD / Admin tabs, EOD commit, identity math, cache layer, mailer, WhatsApp worker, or Ollama integration.

New role-locked portals:

- ? Logistics Portal - 8 tabs (Incoming PRs - Create PO - Open POs - Assign to Warehouse - Reschedules - Force-Close - Vendor Returns - History). Documented in §14.
- ? Warehouse Portal - 6 tabs (Incoming Assignments - Receive Goods - Prepare DN - Outbound DNs - Returns from Site - History). Documented in §15.

New tabs / expanders on existing pages:

- HOD Portal -> ? DN Approvals (§6.15) and ? In-Transit (§6.16) - 14th and 15th HOD tabs.
- HOD Portal -> Purchase Requests -> new "? Submit PR(s) to Logistics Portal" expander (§6.8.4a). Coexists with the existing email path.
- SK Entry Log -> Receipt Staging -> new "? Incoming Delivery Notes from Warehouse" expander (§4.4.0).
- Admin Portal -> ? Logistics Oversight (§7.11a) - 11th admin tab.

New sidebar component:

- Notifications bell with unread badge + inbox dialog (§3.6).

New background job:

- T-2 / T-1 / T-0 delivery reminders fired by `whatsapp_worker._maybe_run_delivery_reminders()` once per local day (§9.4).

New reports: PO Status / Warehouse Throughput / Force-Closures (added to the existing Reports module, available to Supervisor + HOD + Admin).

RL/BL strict separation: Rubber Lining and Brick Lining never aggregate. Enforced in `po_items.rl_bl_family` tagging, DN splitter rejection, and DN header family tag. See §11.4e.

Warehouse-blind pricing: Three independent enforcement layers ensure Warehouse users never see `Unit_Price`, `Total_Price`, or any monetary header field. See §15.

Bug fixes that shipped with v3.0:

- Double GMT+3 addition removed from Admin Pending Requests, HOD My Requests, Admin WhatsApp Console (`_localize()` already converts at the data-layer boundary).
- Admin Pending Requests now joins inventory to display `Material_Code` + `Material_Name` + UOM alongside `SAP_Code`.
- Sidebar Hub Assistant gracefully handles unreachable Ollama with `st.warning(" Local AI is offline. Please run 'ollama serve'...")`.

14. Logistics Portal Manual

The Logistics Portal sits between Site HOD (who creates PRs) and Warehouse (which physically receives goods). Role-locked to {logistics, admin} - exact-role lock means no other role inherits access via the hierarchy. Eight tabs.

14.1 Pages visible to Logistics

- Live Dashboard (read-only - all sites)
- ? Logistics Portal (this page)
- Reports (incl. the 3 new procurement reports)

14.2 Tab 1: Incoming PRs

Site HODs submit PRs to Logistics from their HOD Portal -> PR tab. They land here.

14.2.1 Site filter

Dropdown of all sites + "All sites". Defaults to all.

14.2.2 Hero strip (3 cards)

Card	Source
OPEN PRs	Active queue count
TOTAL QTY	Sum across all open PR lines
SITES	Distinct sites currently submitting

14.2.3 Queue table

AgGrid with columns: PR No. - Site - Lines - Total Qty - Submitted - Earliest Delivery - Status. Sortable, filterable, exportable.

14.2.4 Per-PR drilldown

Selectbox under the queue table. Pick a PR + Site combination and a section card appears:

Card row	Source
PR Number	the row
Site	the row
Lines	line count
Total Qty	summed across lines
Earliest Delivery	min of Delivery_Date across lines

Below: full line items grid with Material_Code, Material_Name, Requested_Qty, UOM, WBS_Number, Network, Plant, Delivery_Date, Supplier, Est_Cost_SAR.

14.2.5 ? Use this PR to create a PO button

Loads the PR into Tab 2 (Create PO). Selectbox state is preserved so you can switch tabs without losing the chosen PR.

14.3 Tab 2: ? Create PO

Two sub-tabs: ?? Manual entry and ? PDF upload. Both funnel into the same purchase_orders insert path.

14.3.1 Vendor picker

Selectbox of all active vendors `<code> - <name>`, plus a "? Add new vendor" option that opens an inline expander with Vendor Code, Name, Address, Default Inco Terms, Default Payment Terms. Save -> re-renders the parent form with the new vendor pre-selected.

14.3.2 PO header form

Three-column layout:

Column	Fields
Left	PO Number * - PO Date - PO Type
Middle	PR Number - Quotation No. - Quotation Date
Right	Expected Delivery - Inco Terms - Payment Terms

Plus a second three-column row for Contact (vendor), Contact Email, Mobile, Our Reference, Your Reference, Our Email.

Vendor defaults (Inco / Payment Terms) auto-fill from the master row but remain editable.

14.3.3 Manual sub-tab: line items

If a PR was loaded via the Tab 1 shortcut, every PR line pre-fills with Include = True and zero Unit_Price + Total_Price (you set them). Otherwise an empty editable grid opens.

Editable columns: Include - Material_Code - Description - Qty - UOM - Unit_Price - Total_Price - WBS_Number - Network - Plant. Rows with Include = False are dropped on save.

14.3.4 PDF upload sub-tab

Element	Purpose
File uploader	Drop the PO PDF
? Extract from PDF button	Calls process_po_pdf() - regex-based extraction of header (PO Number, Vendor, Inco/Payment, Quotation refs, totals) + line items (Sr. No, Material_Code, Description, Qty, UOM, Unit_Price, Total_P...
Review extracted PO	Editable preview of every extracted field. The header form pre-fills. The line items grid pre-fills. Edit anything before saving.

Element	Purpose
Delivery schedule	If Annexure parsed, a table shows shipment_no - material_group - target_date
Save PO (from PDF)	Persists the PO + items + shipment schedule. The original PDF is stored as a BLOB on the purchase_orders row (attachment_blob/_name/_mime) for audit.

On PO numbers with X-masking: the sample PDF has the last 4 digits masked as XXXX for security. The extractor preserves whatever is on the page verbatim. In production, vendors send full 10-digit numbers and those pass through unchanged.

14.3.5 Side-effects on save

- New row in purchase_orders with status 'open'
- One row per item in po_items with rl_bl_family auto-tagged
- Linked PR rows (matching PR_Number + Site_ID) flip to logistics_status='in_po' so they leave your Incoming PRs queue
- Site HOD gets po_issued notification (in-app + WhatsApp gated)

14.4 Tab 3: Open POs

Browse + drill into every open PO.

14.4.1 Filters

Site dropdown - Vendor dropdown - PR Number exact-match textbox.

14.4.2 KPI strip

OPEN POs count - Total value SAR - PENDING (not yet delivered) count.

14.4.3 Per-PO drilldown

Select a PO -> section card with PR Number, Vendor, Inco/Payment Terms, PO Date, Expected, Source (manual/pdf_upload), Status, Total. Below: items grid with an RL/BL family chip per row, the parsed delivery schedule (if any), and a list of warehouse assignments.

14.5 Tab 4: ? Assign to Warehouse

Route a PO (or a subset of items) to a warehouse for receiving.

14.5.1 PO + Warehouse pickers

Two-column row: PO selectbox - Warehouse selectbox (shows <id> - <name>).

14.5.2 Items selector

Editable grid showing every PO line with an Include checkbox (default True). Disable lines you don't want to route - they stay with this PO for a future assignment.

14.5.3 Header

Expected Delivery date + Notes (visible to Warehouse).

14.5.4 Assign to Warehouse button

- Subset = if all items included, encoded as None (cleaner audit)
- Subset = if partial, encoded as JSON list of po_items.id in po_assignments.items_subset_json
- Warehouse user(s) at the chosen WH get a po_assigned_to_warehouse notification (in-app + gated WhatsApp)
- PO header's Expected_Delivery auto-fills with this date if not set
- Audit log entry: ASSIGN_PO_TO_WAREHOUSE

14.6 Tab 5: ? Reschedules

Incoming reschedule requests from Warehouse / Site HOD.

14.6.1 Empty state

If no pending requests: empty-state card.

14.6.2 Per-request card

Element	Purpose
Header line	PO <number> - DN <number> - From <current_date> -> <requested_date>
Subline	requested_by + role + reason
Decide popover	Decision notes textbox + [OK] Approve / [X] Reject buttons (reject requires reason)

Approval auto-pushes the new date to: the PO header Expected_Delivery, the po_assignments.Expected_Delivery, and the DN DN_Date (if linked).

14.7 Tab 6: ? Force-Close

Use sparingly. Force-closing notifies Admin + originating Site HOD immediately with the reason.

14.7.1 Target radio

PR (entire) - PO (entire) - PO line (single item)

14.7.2 Reason textbox

Mandatory, minimum 3 characters. Logged verbatim to po_force_closures + audit.

14.7.3 Target picker (changes per radio)

- PR: dropdown of open PR numbers in Logistics queue
- PO: dropdown of open POs
- Line: PO dropdown -> then line dropdown showing only open / partially_delivered lines

14.7.4 ? Force-close button

Single confirm-and-execute. Behind it:

- pr_master.status='closed' + logistics_status='force_closed' (for PR target)
- purchase_orders.status='force_closed' + line statuses for force_closed (for PO target)
- po_items.line_status='force_closed' (for line target)
- New row in po_force_closures with reason
- Notifications fan-out: Admin (recipient_role='admin') + originating Site HOD (recipient_role='hod', recipient_site=<site>), severity = critical

14.7.5 Recent force-closures (audit)

AgGrid of the last 50 closures. Read-only.

14.8 Tab 7: <-? Vendor Returns

Raise a return to the vendor against a PO. Returning a line REOPENS the PO so it shows in your active queue again.

14.8.1 PO picker + scope

PO selectbox (all POs incl. closed). Then radio: Whole PO or Single line. If single line, a line picker appears.

14.8.2 Return details form

- Return quantity (number, > 0)
- Reason (textarea, mandatory)
- Expected resupply (date picker, defaults to today + 14)
- Notes (optional)

14.8.3 <-? Raise vendor return button

- New row in po_returns with raised_by_role='logistics'
- po_items.Returned_Qty bumps; line_status flips back to partially_delivered or open
- If the PO had been closed, header flips back to partially_delivered and closed_by / closed_at / close_reason are nulled
- Notification: vendor_return_raised to logistics inbox + WhatsApp gated

14.8.4 Open returns table

AgGrid of all open returns. Read-only.

14.9 Tab 8: History

Read-only archive. Two sub-tabs: Closed POs and Closed PRs.

- Closed POs: status in closed / force_closed / cancelled
- Closed PRs: any PR where pr_status='closed' OR logistics_status in force_closed/closed/in_po

For force-closures with the reason history, use Tab 6 -> Recent force-closures table.

14.10 Logistics - Use Cases

Use Case 1: Issue PO from a fresh PR

1. Incoming PRs -> pick PR -> ? Use this PR to create a PO
2. ? Create PO -> ?? Manual entry -> pick / add vendor, fill header, edit Unit Price per line
3. Save PO -> balloon animation -> notification fires to Site HOD

Use Case 2: Issue PO from a vendor's emailed PDF

1. Receive PO PDF from vendor (sample format: ? Create PO -> ? PDF upload)
2. Drop the PDF -> ? Extract from PDF
3. Review extracted header + line items + delivery schedule
4. Correct anything the parser missed
5. Save PO (from PDF) - original PDF is archived

Use Case 3: Route a PO to a warehouse

1. ? Assign to Warehouse -> pick PO + warehouse
2. Disable lines you want to route separately later
3. Pick Expected Delivery, add a routing note
4. Assign - warehouse user is pinged

Use Case 4: Handle a Warehouse reschedule ask

1. ? Reschedules -> review the request (reason text + current vs requested date)
2. Decide -> notes + [OK] Approve
3. PO Expected_Delivery auto-updates; Warehouse + requester get the decision notification

Use Case 5: Force-close a stale PR

1. ? Force-Close -> radio = PR (entire)
2. Write the reason (e.g. "Project cancelled by site management 2026-06-15")
3. Pick PR -> ? Force-close PR
4. Admin + Site HOD see a critical notification in their bell

14.11 Logistics - FAQ

Q: A PR appears in Incoming PRs but I can't issue a PO against it - Material_Code is wrong. A: PR lines come from the Site HOD's inventory catalogue. If the Material_Code is wrong, the SK created it wrong upstream. Don't fix in your PO - bounce the PR back via WhatsApp / chat, ask the site to re-submit. There's no in-app "reject PR" button (yet) because PR rejection should be a conversation, not a click.

Q: PDF extraction picked up wrong qty. A: The Review preview is editable. Always check Qty / Unit_Price / Total_Price before saving. The extractor is calibrated against the General Industries sample layout; other vendor templates may need template additions.

Q: I want to assign a PO to multiple warehouses. A: Two separate ? Assign actions - one per warehouse. The items_subset_json field on each assignment row keeps them distinct.

Q: Why can't I see prices in Tab 4 / on assignment cards? A: You CAN see prices in Tab 3 (Open POs drilldown). Tab 4 (Assign to Warehouse) shares the read with Warehouse users, so prices are hidden there for consistency.

Q: Force-closure undo? A: Not yet. Once force-closed, you'd have to either raise a Vendor Return (reopens the line) or admin-edit via Master DB Editor. A 24-hour undo window is on the v3.0 backlog.

15. Warehouse Portal Manual

The Warehouse Portal is the physical-receiving and DN-preparation side. Role-locked to {warehouse_user, admin}. Six tabs. Prices are completely hidden in every view - three independent enforcement layers guarantee Unit_Price, Total_Price, Total_Amount, Freight_Charges, Handling_Charges, Discount_Amount, Amount_In_Words are never visible to a warehouse user.

15.1 Pages visible to Warehouse User

- Live Dashboard
- ? Warehouse Portal (this page)
- Reports

15.2 Sidebar warehouse resolution

- A warehouse_user is bound to a single warehouse via users.Warehouse_ID. The portal auto-resolves from the user's profile.
- An admin shadowing the portal gets a sidebar "? Shadow warehouse" selectbox listing every active warehouse.
- If neither resolves: red error card "? Your account is not bound to a warehouse. Ask Admin to set your Warehouse_ID in Admin Portal -> Users."

15.3 Page title

? Warehouse <ID> in gold, with (prices hidden - Logistics-only) muted caption next to it as a permanent visual reminder.

15.4 Tab 1: Incoming Assignments

POs Logistics has routed to this warehouse.

15.4.1 Hero strip

Card	Source
AWAITING ACK	Count of assignments with status='assigned'
ACKED	Status = acknowledged (you've seen them, waiting on goods)
IN/RECEIVED	Status in partial / received

15.4.2 Assignments grid

AgGrid: Assign # - PO No. - PR No. - Vendor - Dest Site - Expected - Assigned - Acked - Status. No price columns.

15.4.3 Acknowledge action

Selectbox of status='assigned' rows. Picking one -> [OK] Acknowledge button -> flips to acknowledged and pings Logistics with warehouse_acknowledged notification (off by default in WhatsApp toggles to keep noise low).

15.5 Tab 2: Receive Goods

Record qty actually received at this warehouse against an acknowledged assignment.

15.5.1 Assignment picker

Selectbox of status IN ('acknowledged', 'partial') rows.

15.5.2 PO snapshot card (no prices)

KV row	Value
PO Number	the PO
Vendor	<code> - <name>
Inco / Payment	the terms
Expected	the date

Total_Amount, Freight_Charges, Handling_Charges, Discount_Amount, Amount_In_Words are stripped from the header dict before render. Never visible.

15.5.3 Receive grid

Editable grid showing every line on the assignment:

Column	Source
id	po_items.id (disabled - read-only)
Material_Code	line code
Description	line description
UOM	unit
rl_bl_family	family chip (RL / BL / blank)
Qty	ordered (read-only)
Delivered_Qty	already received cumulative (read-only)
Open_Qty	computed Qty ? Delivered + Returned (read-only)
Receive Now	EDITABLE - type the qty you physically received this event

15.5.4 Record receipt button

- Validates: at least one row with Receive Now > 0
- Over-deliver guard: Delivered_Qty ? Returned_Qty + new_qty may NOT exceed ordered Qty. If it does, the entire batch is rejected with a friendly message naming the line.
- On success: bumps each line's Delivered_Qty, flips line_status (delivered if Delivered ? Returned ? Qty, else partially_delivered)

—

Rolls assignment status (received if every line on the parent PO is delivered, else partial)

- Rolls PO header status to match
- Notification: warehouse_received to Logistics with line count

15.6 Tab 3: Prepare DN

Build a Delivery Note for a site. RL/BL strict separation is enforced here - a DN cannot span both families. If you try, the action is rejected with: **"Strict separation violated: this DN spans multiple RL/BL families. Prepare one DN per family."**

15.6.1 Source pickers

- PO No. selectbox (assignments with status acknowledged / partial / received)
- Destination Site selectbox

15.6.2 Items grid

Editable grid with read-only inventory columns (Material_Code, Description, UOM, rl_bl_family, Qty, Delivered_Qty, Returned_Qty) and editable shipping columns:

Column	Purpose
Ship Qty	What you're sending on this DN (must be > 0 to include)
Lot_Number	If you tracked the lot at receive time
Expiry_Date	Lot expiry
Remarks	Free text

15.6.3 DN header

Three-column row: DN Date - Vehicle No - Driver Name - Driver Phone - Prepared By (auto-filled with your username) - Remarks.

15.6.4 Save DN draft button

- Validates: at least one row with Ship Qty > 0
- Over-ship guard: available = Delivered_Qty ? Returned_Qty ? (live DN qty) per line; ship qty cannot exceed available
- RL/BL strict-separation check: if the items span both families, REJECTED
- On success: new delivery_notes row with status='draft' and one dn_items row per line. The DN header carries rl_bl_family if non-NULL.
- Toast: DN <number>

15.6.5 DN numbering convention

DN-<WAREHOUSE_ID>-<YYYYMMDD>-<seq> - seq resets per (warehouse, day). Example: DN-WH-A-20260616-003.

15.7 Tab 4: ?? Outbound DNs

Track every DN this warehouse has prepared.

15.7.1 Status filter

Multi-select of every DN status. Defaults to draft, pending_logistics, pending_hod, pending_sk (the active ones).

15.7.2 DN grid

AgGrid of every matching DN with full state metadata.

15.7.3 Submit-to-Logistics action

When the filter shows any draft row, a "Submit a draft to Logistics" section appears with a selectbox of drafts + Submit to Logistics button. Submission flips status to pending_logistics and pings Logistics with the DN-approval-queue notification.

15.7.4 Per-DN drilldown

Select a DN -> section card showing PO, Site, Status, RL/BL family, Vehicle/Driver, and the full signature trail (Logistics decided by/decision, HOD decided by, SK received by). Items grid below.

15.7.5 ? Request reschedule expander

If the date you targeted isn't going to work, raise a reschedule from here. Same flow as the HOD's In-Transit tab (defaults, min_value=today, mandatory reason).

15.8 Tab 5: <-? Returns from Site

Raise a return when a site flags defective material from a DN this warehouse delivered.

15.8.1 DN picker

Selectbox of DNs in received / pending_sk / hod_approved states.

15.8.2 Items grid

Editable grid with a Return Qty column. Filling > 0 on a line includes it in the return.

15.8.3 Reason textarea

Mandatory.

15.8.4 <-? Raise return to vendor button

- Internally calls record_internal_return() which fans out to raise_vendor_return() per affected line
- Each line: po_returns row written with raised_by_role='warehouse_user', dn_item flagged returned,

parent po_item's Returned_Qty bumps and line_status flips back to partially_delivered or open

- PO header reopens if it had been closed
- Notification fires to Logistics

15.9 Tab 6: History

Read-only. Two sub-tabs: Completed DNs (status in received / rejected / cancelled) and Closed assignments.

15.10 Warehouse - Use Cases

Use Case 1: Receive a vendor delivery

1. Incoming Assignments -> [OK] Acknowledge the assignment when you see it
2. Goods physically arrive
3. Receive Goods -> pick the assignment -> type the actual received qty per line -> Record receipt

Use Case 2: Ship to a site (single family)

1. Prepare DN -> pick PO + destination site
2. Type Ship Qty per line - keep all RL or all BL
3. Fill DN header (Vehicle, Driver, Date)
4. Save DN draft -> status = draft
5. ?? Outbound DNs -> select the draft -> Submit to Logistics

Use Case 3: Ship RL + BL to the same site on the same day

You need TWO DNs:

1. Prepare DN with only RL lines -> save + submit
2. Prepare second DN with only BL lines -> save + submit

The system rejects any attempt to combine them on a single DN.

Use Case 4: A site rejects part of a DN - raise to vendor

1. <-? Returns from Site -> pick the DN
2. Type Return Qty on the offending line
3. Reason: "Site reported defective surface coating"
4. <-? Raise return to vendor - PO + po_item reopen; Logistics sees the return in their tab

Use Case 5: Request a reschedule

1. ?? Outbound DNs -> drill into the affected DN
2. ? Request reschedule for this DN -> pick new date + reason -> Submit

15.11 Warehouse - FAQ

Q: Why don't I see prices anywhere? A: Warehouse role is intentionally blind to commercial data. Three independent enforcement layers strip every monetary field. If you genuinely need a price (e.g. to file a damage claim), ask Logistics or Admin.

Q: My over-ship guard rejected a DN. I just received everything from the vendor. A: Probably another DN is already drafted (or submitted) shipping a slice of that line. Check ?? Outbound DNs filter for live DNs against the same PO. If one was abandoned, ask Logistics to reject/cancel it so the qty frees back up.

Q: RL and BL are going to the same site on the same truck. Can I combine? A: No. Two DNs. The strict separation is by design (different testing standards, different storage requirements). The site receives them as two separate DNs in their Receipt Staging queue.

Q: Where did the "Receive Goods" assignment go? It was here this morning. A: Once every line on the parent PO is fully delivered, the assignment moves from partial to received and stops appearing in Tab 2 (which filters to acknowledged + partial). Check the assignment grid in Tab 1.

Q: Can I edit a DN after submitting to Logistics? A: Not directly. Ask Logistics to reject it (?? Outbound DNs shows the rejection if it happens) - then it returns to draft for you to edit.

16. Cross-Role Procurement Walk-through

A single happy-path narrative threading all five roles together, useful for onboarding.

16.1 The scenario

Site GI-PS01 needs 50 RL panels and 20 BL bricks. Total 3 working days from PR raise to physical receipt.

16.2 Hour by hour

When	Who	What
Day 1 - 8:00 AM	Site HOD	Opens HOD Portal -> Purchase Requests. Creates 2 PR lines (50 RL panels, 20 BL bricks) with PR Number 3000099999, WBS Number
Day 1 - 8:15 AM	Site HOD	Opens the ? Submit PR(s) to Logistics Portal expander -> multi-selects PR 3000099999 -> Submit Selected to Logistics.
Day 1 - 8:15 AM	(auto)	pr_submitted_to_logistics notification -> Logistics inbox (red badge). WhatsApp ping if enabled.
Day 1 - 9:00 AM	Logistics	Sees PR 3000099999 in Incoming PRs queue. Drills in - 2 lines, RL + BL. Clicks ? Use this PR to create a PO.
Day 1 - 9:30 AM	Logistics	Switches to ? Create PO -> manual entry. Picks vendor "Carborundum Universal" (0000110341), Inco/Payment auto-fill, fills PO Nu...
Day 1 - 9:30 AM	(auto)	PR rows flip to logistics_status='in_po' -> disappear from Logistics queue. Site HOD gets po_issued notification with vendor +
Day 1 - 10:00 AM	Logistics	? Assign to Warehouse -> picks PO 4720033030 + Warehouse WH-A + Expected Delivery Day 2 evening. Assign.
Day 1 - 10:00 AM	(auto)	po_assigned_to_warehouse notification -> WH-A users' inbox. WhatsApp ping if enabled.
Day 1 - 11:00 AM	Warehouse user (WH-A)	Incoming Assignments -> [OK]

Acknowledge assignment #N. Goes about their day.

When	Who	What
Day 2 - 5:00 PM	Warehouse user	Physical truck arrives from vendor. Receive Goods -> assignment #N -> type 50 in Receive Now for the RL line, 20 for the BL li...
Day 2 - 5:30 PM	Warehouse user	Prepare DN -> pick PO 4720033030, destination GI-PS01. Types Ship Qty 50 for RL line, attempts to add BL line - rejected (RL/B...
Day 2 - 5:35 PM	Warehouse user	Repeats: new DN with only the BL line -> DN-WH-A-20260617-002.
Day 2 - 5:40 PM	Warehouse user	?? Outbound DNs -> submits both drafts to Logistics.
Day 2 - 5:45 PM	Logistics	Sees DN-WH-A-20260617-001 + 002 in their DN approval queue. Confirms delivery date Day 3 AM. Approves both.
Day 2 - 5:45 PM	(auto)	DN status flips to pending_hod. Site HOD gets dn_logistics_approved notification.
Day 2 - 6:00 PM	Site HOD	Logs in, sees red bell badge 2 unread. Opens inbox -> drills into the DNs via ? DN Approvals tab. Approves both.
Day 2 - 6:00 PM	(auto)	DN status pending_sk. Two mirror rows in pending_receipts.status='pending_sk'. Site SK gets dn_auto_generated notification.
Day 3 - 8:30 AM	Truck arrives at site GI-PS01	
Day 3 - 8:45 AM	Site SK	Opens Entry Log -> Receipt Staging. The new ? Incoming Delivery Notes from Warehouse expander is open at the top, showing both...
Day 3 - 8:45 AM	(auto)	Two receipts rows written (with DN_Number, Warehouse_ID, PO_Number_Source). Both DNs flip to received. pending_receipts mirror

...

16.3 Side-path examples

A site rejects a DN

Same scenario, but on Day 2 at 6:00 PM, Site HOD inspects the line preview on DN-002 and notices the BL qty doesn't match the PR. Rejects DN-002 with note "BL qty in DN is 25, PR asked for 20 - bounce back".

—

DN-002 status -> rejected, rejection_reason stored

- Warehouse user gets the rejection notification with the reason
- The BL line on the PO frees up qty (the over-ship guard recalculates available since DN-002 is no longer "live")
- Warehouse prepares DN-003 with correct qty

Logistics force-closes the PR mid-flight

Project cancelled by management on Day 1 evening:

- Logistics -> ? Force-Close -> radio = PR -> reason "Project cancelled per CFO email 2026-06-16" -> ? Force-close PR
- Admin + Site HOD see critical-severity notification in their bell
- The pending PO can either stay (if vendor already shipped) or also be force-closed
- Site HOD's ? In-Transit -> Force-closures sub-tab shows the closure with the reason

Warehouse asks for reschedule

Receiving day Truck doesn't show up - vendor delay:

- Warehouse -> ?? Outbound DNs -> drill into DN-001 -> ? Request reschedule -> new date Day 4 -> reason "Vendor delivery delayed 24h confirmed by vendor email"
- Logistics -> ? Reschedules -> reviews -> [OK] Approve with notes "Confirmed with vendor"
- PO Expected_Delivery, po_assignments.Expected_Delivery, AND DN_Date all flip to Day 4 in one transaction
- Site HOD sees the decision notification

17. Operations & Hosting - the after-launch chapter

This chapter is what the host operator (you, today: johnsonandrew) needs every day after the system is live. Everything before this chapter is about the application; this chapter is about keeping the application running on a real Mac that real users hit through <https://gi.giinventory.com>.

If you're reading this and you haven't deployed yet, do §14.2 First-time setup straight through, then come back to this chapter as needed.

14.1 The system in one diagram

```

????????????????????????????
? User on phone / laptop      ?
? (any @generalindustries    ?
? .net email)                 ?
????????????????????????????
      ? HTTPS (TLS 1.3)
      ?
????????????????????????????
? Cloudflare Edge             ?
? * Access email allow-list?
? * DDoS + WAF                ?
? * Cert auto-renewal         ?
????????????????????????????
      ? encrypted tunnel
      ? (outbound from Mac, no open port)
      ?
????????????????????????????
? Your Mac (johnsonandrew, FileVault on)      ?
?                                              ?
? ?? launchd ?????????????????????????????? ?
? ?  com.gi.streamlit      -> Streamlit on :8501  ? ?
? ?  com.gi.whatsapp-worker -> drives Chrome / WA Web? ?
? ?  com.gi.cloudflare     -> outbound tunnel    ? ?
? ?  com.gi.backup         -> nightly @ 02:00    ? ?
? ????????????????????????????????????????? ?
?                                              ?
? ?? files ????????????????????????????????? ?
? ?  gi_database.db        (sqlite, FileVault-   ? ?
? ?                        encrypted at rest)    ? ?
? ?  uploads/<Site>/<doc>/   (BLOBs mirrored)     ? ?
? ?  ~/.cloudflared/       (tunnel creds)        ? ?
? ????????????????????????????????????????? ?
?                                              ?
? ?? local services ????????????????????????? ?
? ?  Ollama (qwen2.5-coder, llama3.1, qwen2.5vl) ? ?
? ?  Google Chrome (signed-in WhatsApp Web tab) ? ?
? ????????????????????????????????????????? ?
????????????????????????????????????????
      ?
      ? nightly 02:00
      ?????????????????????????????
      ? iCloud Drive backup      ?
      ? ~/..../GI_Hub_Backups/    ?
      ? * 14-day retention        ?
      ? * DB + uploads/ mirror    ?
      ?????????????????????????

```

No part of this stack stores your inventory data anywhere except the Mac itself. Cloudflare forwards encrypted bytes. Apple stores encrypted backups. The application database lives on a FileVault-encrypted disk on your machine, in your office.

14.2 First-time setup (compressed checklist)

The full step-by-step is in `host_setup/README.md`. Compressed checklist for cross-referencing:

1. Install cloudflared (binary download, no Brew):

```
sudo rm -f /usr/local/bin/cloudflared
cd ~/Downloads
curl -L -o cloudflared.tgz https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared.tgz
tar -xzf cloudflared.tgz
sudo mv cloudflared /usr/local/bin/cloudflared
sudo chmod +x /usr/local/bin/cloudflared
```

1. Create the tunnel:

```
cloudflared tunnel login          # opens browser -> pick giinventory.com
cloudflared tunnel create gi-hub  # prints UUID - copy it
cloudflared tunnel route dns gi-hub gi.giinventory.com
```

1. Write tunnel config: copy `host_setup/cloudflared_config.yml.example` to `~/cloudflared/config.yml` and substitute the UUID + your username.
2. Install launchd services:

```
cd "/Users/johnsonandrew/Downloads/CNCEC PROJECT"
./host_setup/scripts/install.sh
./host_setup/scripts/install.sh --status
```

All four should be [OK] green (backup is ? yellow until 02:00 - that's correct).

1. Set up Cloudflare Access (email allow-list - see §14.5 for details and rationale):
 - Cloudflare -> Zero Trust -> Access -> Applications -> Add -> Self-hosted
 - Domain: `gi.giinventory.com`
 - Policy -> Action: Allow -> Include: Emails ending in `@generalindustries.net`
5. Pair WhatsApp Web on the host Mac (Chrome recommended; Safari works too):
 - Open `https://web.whatsapp.com` -> scan QR with phone -> Pin Tab.
7. Hardening (mandatory before sharing the URL):
 - FileVault on: System Settings -> Privacy & Security -> FileVault -> On.
 - Change every default password in the app (User Management).
 - Screensaver off / no lock: System Settings -> Screen Saver -> Start after = Never.
 - No automatic sleep on power: System Settings -> Battery -> Prevent automatic sleeping = On.
12. Pull a baseline backup before letting anyone in:

```
./host_setup/scripts/backup_db.sh
```

14.3 Daily operations - the 4 commands you'll actually use

Need	Command
Show service status	<code>./host_setup/scripts/install.sh --status</code>
Stream all four log files	<code>./host_setup/scripts/install.sh --logs</code>
One-shot backup right now	<code>./host_setup/scripts/backup_db.sh</code>
Restart Streamlit after git pull	<code>./host_setup/scripts/restart_app.sh</code>

Logs live in `~/Library/Logs/`:

- `gi-streamlit.{log,err}`
- `gi-whatsapp-worker.{log,err}`
- `gi-cloudflared.{log,err}`
- `gi-backup.{log,err}`

14.4 Updating the host with new code or features

This is the most common operation. Average downtime: ~3 seconds.

```
cd "/Users/johnsonandrew/Downloads/CNCEC PROJECT"

# 1. Pull the new code
git pull

# 2. If requirements.txt changed (added/upgraded a Python package)
.venv/bin/pip install -r requirements.txt

# 3. Restart Streamlit only
./host_setup/scripts/restart_app.sh

# If worker code (whatsapp_worker.py) changed too:
./host_setup/scripts/restart_app.sh --worker

# If you also changed cloudflared or backup config (rare):
./host_setup/scripts/restart_app.sh --all

# 4. Verify everything came back
./host_setup/scripts/install.sh --status
```

`init_db()` runs on every Streamlit start, so any new schema columns or tables in the pulled commit are auto-applied before any user can sign in. You never run SQL migrations by hand.

For schema changes that need backfilling (rare), the relevant `init_db()` block also handles the backfill - those are clearly commented "one-time backfill" in `database.py`.

Rolling back if a release breaks something

```
git log --oneline -5                                # find the last-known-good commit hash
git checkout <good-commit-hash>
./host_setup/scripts/restart_app.sh
```

Your database has not been migrated backwards because schema changes are additive (new columns / tables - old columns are never dropped). Old code can read the newer schema fine.

14.5 Security & data safety - what to tell management

The full audit-quality version is in handoff.md. The customer-friendly version follows.

Where does our data physically live?

- On the Mac in our office. The database file (gi_database.db) and every uploaded attachment (delivery notes, MTC certificates, photos) sit on the Mac's own disk. Nothing is uploaded to a third-party database service.
- FileVault encryption scrambles the entire disk. A stolen Mac is a brick - the database is unreadable without the disk password.
- Encrypted backups to iCloud Drive add a second copy at Apple, also encrypted, also under your control.

Who can reach the application?

- Only the email addresses we explicitly whitelist. Cloudflare Access checks the user's email at the edge - anyone NOT on the list never reaches the Mac, never sees the login page, never causes a single log line on our server. The list is one-click manageable in the Cloudflare dashboard.
- And inside the app: users still sign in with the role-based bcrypt login. Two locks on the door, not one.

How is the connection secured?

- Every byte travels over HTTPS (TLS 1.3 - the same encryption layer used by banks).
- Cloudflare forwards encrypted bytes; they cannot read the contents. (They terminate TLS at their edge, then re-encrypt to our Mac via the Tunnel's own mTLS channel. The application body is opaque to them.)
- Cloudflare certifications relevant to this: SOC 2 Type II, ISO 27001, PCI DSS. Public reference: <https://www.cloudflare.com/trust-hub/>

What's the attack surface?

- Zero public ports. Our Mac never accepts an inbound TCP connection. The Tunnel is an outbound connection from us to Cloudflare. A hacker port-scanning the internet sees nothing.
- The application gates every consequential action behind a role check (HOD/Admin/etc.). Even an authenticated user can't escalate privileges from the UI.
- Every action writes a permanent audit log row (system_audit_log) with username + timestamp + details. Admin can review or export at any time.

What happens if X fails?

Failure	Impact	Recovery
Power outage	App offline for the duration	UPS battery (recommended, ~\$50) keeps it up; otherwise it restarts on

power-on

Failure	Impact	Recovery
Internet outage	External users blocked; office network still reaches <code>http://localhost:8501</code>	None needed - auto-resumes when ISP returns
Cloudflare incident	External access blocked	Switch to Tailscale Funnel in ~5 minutes (documented in <code>handoff.md</code>)
Host Mac dies	App down until replacement	Restore last backup to a new Mac, ~30 min, lose ~24h of data
Disk corruption	DB unusable	Restore from a backup (14 days kept), ~5 min
Accidental delete	Single record gone	Restore from a backup, ~5 min
Internal user steals data	Limited scope (own role)	Audit log shows username + exact action + timestamp for investigation

What's the actual cost?

- Domain (`giinventory.com`): ~\$8 / year on Cloudflare Registrar (at-cost).
- Cloudflare Tunnel: free, unlimited bandwidth.
- Cloudflare Access ~50 users: free.
- iCloud storage (5 GB free, enough for this app): free.
- WhatsApp notifications via WhatsApp Web on the host Mac: free.
- Local Ollama AI models: free, computed on the Mac.
- Mac electricity: ~\$50-80 / year.

Total: \$58-88 / year, all-in.

Non-negotiables for the operator

These are not optional. Skipping any of them defeats the security model.

1. FileVault must be on.
2. The host Mac stays in a physically secured location (locked IT room or office).
3. Default passwords (`admin/admin2026` etc.) must be changed on day one.
4. The Cloudflare Access policy must be set BEFORE the URL is shared with any user.
5. The Mac must not be used for personal browsing. Keep it as a single-purpose appliance.
6. Apple Software Updates for macOS get installed within 7 days of release.

14.6 Backup & restore

Automatic nightly

`com.gi.backup` fires at 02:00 every day via `StartCalendarInterval` in the `launchd` plist. The script (`host_setup/scripts/backup_db.sh`) does:

1. SQLite online backup to ~/Library/Mobile

Documents/com~apple~CloudDocs/GI_Hub_Backups/db/gi_database_YYYYMMDD_HHMMSS.db.

The online backup API takes a consistent snapshot even while Streamlit is reading + writing, with no downtime.

2. rsync mirror of uploads/ to ~/.GI_Hub_Backups/uploads_latest/ (incremental - only changed files transfer).
3. Prune snapshots older than 14 days.
4. Optional second destination if you've set the env var GI_BACKUP_EXTRA (e.g. an external SSD path or rclone-mounted Google Drive).

Verify the most recent backup any time:

```
ls -lht ~/Library/Mobile\ Documents/com~apple~CloudDocs/GI_Hub_Backups/db/ | head -5
```

Manual snapshot before risky work

Always run a snapshot before:

- Major version upgrade (git pull that touches database.py)
- Bulk edits via Admin -> Master DB Editor
- Switching the host Mac to a new machine
- Migrating to a different hosting path

```
./host_setup/scripts/backup_db.sh
```

Restore

1. Stop the app first (the live DB must not be open during a restore)

```
launchctl unload ~/Library/LaunchAgents/com.gi.streamlit.plist
```

2. Copy the chosen snapshot into place

```
cp "~/Library/Mobile Documents/com~apple~CloudDocs/GI_Hub_Backups/db/gi_database_20260613_020000.c"
"/Users/johnsonandrew/Downloads/CNCEC PROJECT/gi_database.db"
```

3. Start the app again

```
launchctl load -w ~/Library/LaunchAgents/com.gi.streamlit.plist
```

```
./host_setup/scripts/install.sh --status
```

For uploads/ files, restore from the latest mirror:

```
rsync -a "~/Library/Mobile Documents/com~apple~CloudDocs/GI_Hub_Backups/uploads_latest/" \
"/Users/johnsonandrew/Downloads/CNCEC PROJECT/uploads/"
```

Off-site copy (optional but recommended)

iCloud Drive is one copy on Apple's servers. For a truly off-Apple second copy:

Option	Free quota	One-time setup
Backblaze B2	10 GB free, 1 GB/day egress	15 min - brew install rclone, configure B2 backend, point

GI_BACKUP_EXTRA at a mounted
rclone mount or add a second rclone
sync...

Option	Free quota	One-time setup
Cloudflare R2	10 GB free, no egress fees	20 min - same as B2, point rclone at the R2 endpoint
External SSD	depends on disk	0 min - export GI_BACKUP_EXTRA="/Volumes/Ba
Private GitHub repo	unlimited <100 MB files	5 min - <code>cd /git/hub && git commit && git push step in backup_db.sh</code>

Recommended pair: iCloud (always on, encrypted) + Backblaze B2 (off-Apple, off-Mac, off-Cloudflare).
Three independent failure domains.

14.7 Cloudflare cheatsheet

You don't need to be a Cloudflare expert. The four things you'll actually touch:

Add or remove an allowed email

1. Cloudflare dashboard -> Zero Trust (left sidebar, near bottom)
2. Access -> Applications -> click GI Hub Warehouse
3. Policies -> click GI Staff only (or whatever you named it)
4. Edit the Include rule:
 - To allow specific addresses: switch selector to Emails and add the address comma-separated.
 - To allow a whole domain: keep Emails ending in and edit the value (e.g. @generalindustries.net).
7. Save.

Changes take effect within 60 seconds globally. No need to restart anything on the Mac.

Change the tunnel hostname

```
cloudflared tunnel route dns gi-hub anotherhost.giinventory.com
./host_setup/scripts/restart_app.sh --all
```

Cloudflare automatically updates DNS. The old hostname keeps working until you delete its CNAME.

Check tunnel health from the Cloudflare side

Cloudflare dashboard -> Zero Trust -> Networks -> Tunnels -> gi-hub.

You'll see one or more "Connectors" (one per running cloudflared instance). Each shows latency, packet loss, and uptime. If a Connector is missing or grey, your com.gi.cloudflared service isn't healthy - check `~/Library/Logs/gi-cloudflared.err`.

Add custom block / allow rules

Cloudflare -> giinventory.com -> Security -> WAF. You can:

- Block known bad IPs / countries (not usually necessary - Access does the heavy lifting).
- Rate-limit specific paths (e.g. limit /login to 10 attempts/minute per IP - recommended).

14.8 Troubleshooting cheat sheet

Symptom	Most likely cause	First thing to try
https://gi.giinventory.com shows Cloudflare 502	Streamlit isn't running	<code>./host_setup/scripts/install.sh --status</code> - if Streamlit is ? red, check <code>gi-streamlit.err</code>
https://gi.giinventory.com shows Cloudflare 530	Tunnel isn't connected	<code>tail ~/Library/Logs/gi-cloudflared.err</code> - usually a typo in <code>~/cloudflared/config.yml</code>
Cloudflare Access login page rejects a valid email	Email not on the policy	Zero Trust -> Access -> Applications -> GI Hub -> policy -> confirm address is in the Include list
App loads but WhatsApp messages don't deliver	WhatsApp Web tab closed or signed out	<code>open -a "Google Chrome" "https://web.whatsapp.com"</code> - re-pair, leave the tab pinned
WhatsApp delivers, but Cmd+W AppleScript fails first time	Accessibility permission not granted	macOS prompt should appear; click Allow once. Verify under System Settings -> Privacy & Security -> Accessibility
Backup directory empty	iCloud Drive disabled	System Settings -> Apple Account -> iCloud -> iCloud Drive -> On. Re-run <code>backup_db.sh</code>
AI features dead	Ollama not running	<code>ollama serve</code> (or it's already running - check ollama ps). Confirm via Admin -> Settings -> AI Connection
Timestamps wrong	TZ env var not picked up	<code>launchctl unload ~/Library/LaunchAgents/com.gi.streamlit.plist && launchctl load -w ~/Library/LaunchAgents/com.gi.streamlit.plist</code>
Streamlit exit 126	Permissions on the venv binary	<code>chmod +x .venv/bin/streamlit</code> then <code>./host_setup/scripts/restart_app.sh</code>
Need to free disk fast	Old backups	<code>find ~/Library/Mobile Documents/com~apple~CloudDocs/GI_Hub_Backups/db -name 'gi_database_*.db' -mtime +7 -delete</code>
Want a "full uninstall, keep data"	-	<code>./host_setup/scripts/uninstall.sh</code> - services removed, DB + uploads + backups untouched

If you hit something that's not in this table, capture three things and send to the developer:

```
./host_setup/scripts/install.sh --status > /tmp/status.txt  
tail -200 ~/Library/Logs/gi-streamlit.err > /tmp/streamlit-err.txt  
tail -200 ~/Library/Logs/gi-whatsapp-worker.err > /tmp/worker-err.txt
```

Document end

This manual covers every page, tab, button, table, and field built into the General Industries Hub v3.0 as of the latest commit, including the new Logistics and Warehouse portals and the procurement chain. For technical reference (function signatures, table schemas, full SQL), see `database.py`, `auth.py`, and `pages_internal/*.py` source files. For day-to-day operating procedure across all five roles, see `SOP.md`.

For PDF export: use any markdown-to-PDF converter (Typora, pandoc, marp, or VSCode's "Markdown PDF" extension). Suggested pandoc command:

```
pandoc USER_MANUAL.md -o USER_MANUAL.pdf --pdf-engine=xelatex \  
  --toc --number-sections \  
  -V geometry:margin=2cm -V mainfont="Helvetica" -V monofont="Menlo"
```